A Guide to the Mercedes-Benz Electronic Parts Catalog

MBUSA Parts Technical

May 7, 2007

EPCnet v1.17

Table of Contents

The information in this guide is the latest available at the time of publication, and is believed to be accurate. However, there may be errors or omissions in this document, and features and functions in the application may be added, removed, updated or changed without notice. No warranty or guarantee of fitness is implied or offered. We would be grateful to receive any comments or concerns about this document or the EPCnet application.

1. Introduction

Why should we be concerned with specifying parts correctly?

When you select the right parts, you enable your Service Department to complete the repairs in a timely, accurate and professional way. Just as you expect companies you have business with to keep their promises and provide valuable merchandise or service in trade for your money, Mercedes-Benz owners come to your dealership with expectations of maintenance and repairs carried out professionally and at an agreeable price, and are counting on picking their cars up at the promised time. Providing the correct part the first time and every time helps meet those expectations.

Service Technicians are also expecting you to provide them with the correct part, so they can work as efficiently as possible. Service managers and service advisors are also able to maintain their workshop schedules and customer commitments. Remember that excellent service keeps customers coming back, and having a steady flow of satisfied customers is what keeps everyone in business.

Consider for a moment what happens if you specify and order the wrong part:

- The customer is disappointed and, often, inconvenienced
- The technician loses productive labor time
- The wrong part, if ordered, increases inventory cost and possible obsolescence
- The wrong part, if installed, could create an unsafe condition
- The service advisor has to spend time explaining the error to the customer and in making a second appointment
- The dealership may have to provide a loaner car
- Your dealership's scores on the Mercedes-Benz Service Experience Survey questions about "parts availability" are likely to be lower
- Mercedes-Benz USA and the vendor need to manufacture and ship a part that isn't really needed, taking resources away from more useful efforts
- Customers may decide that Mercedes-Benz isn't a good brand, and purchase their next vehicle from a competitor.

None of these are good. The costs and inefficiencies that accompany poor parts specification are high. This guide is designed to help you enhance the skills necessary to correctly specify Mercedes-Benz parts

Specifying parts correctly the first time is important!

2. Getting Started

The purpose of this guide is to introduce you to your most important tool, which you will use most of every day, the Mercedes-Benz **Electronic Parts Catalog**. This guide discusses only the latest version of Mercedes-Benz EPC, known as EPC net or EWA net. While some of this information applies to earlier versions of EPC, and even to Parts Microfiche, most is specific to EPC net.

This module contains three basic sections:

The Mercedes-Benz Parts System: This section explains the basics of the Mercedes-Benz spare parts numbering and organization system.

The EPC net software application: This section explains the various menus, controls, settings and screens found in EPC net. Having a good command of the application is essential for productivity and efficiency.

Parts Specifying: This section explains how to find part numbers in the EPC. Starting with the basics, it also covers more advanced topics.

After studying this guide, you will have a good understanding of the Electronic Parts Catalog and how to use it effectively. With time you will gain additional experience and knowledge, resulting in greater expertise. Plan on reviewing this guide and the on-line help files frequently, as you work towards mastery of specifying Mercedes-Benz parts.

3. Further Skill Development

The most successful Parts Advisors continue to build their skill levels even after participating in focused training such as that presented in this guide. You already possess many of the important professional skills necessary, however furthering your knowledge will pay valuable benefits to you, your colleagues and your employer.

Skill with using a Personal Computer is valuable when working with applications such as the Electronic Parts Catalog. Skill in using a keyboard and mouse, and recognizing when a computer is malfunctioning, will help ensure quality and accuracy. Understanding the kinds of workplace injuries that can occur as a result of improper use of a computer or improperly adjusted workstation equipment, such as Carpel Tunnel Syndrome, as well as recognizing the warning signs of such injuries, will help you avoid these types of problems.

An understanding of automotive technology will help you communicate more effectively with customers and your colleagues in service. Knowledge of the major systems of an automobile, including how they work, the appearance of various components and how they operate will help you understand the questions being asked of you, visualize the part you seek, and be able to explain any issues if you need to seek assistance.

Karl Friedrich Benz built the first automobile in 1886, and since then Mercedes-Benz has produced hundreds of different models with thousands of variants. While you probably won't be asked to specify parts for a Benz Patentwagen, it is helpful to have an understanding of the standard and optional equipment found in Mercedes-Benz vehicles, going back at least four to six years. This includes common interior and exterior colors, vehicle body styles, model names, the structure of a Vehicle Identification Number (VIN), and Mercedes-Benz Baumusters. This will help you select the correct part, and spot incorrect parts, if not everything about the vehicle's equipment is known.

As you become more familiar with Mercedes-Benz parts, you will also become more familiar with the sometimes unique names used for certain items and systems. Terms like Leather Twin and MB-Tex, Apparatus Case, Frame Floor System, D2B, Aggregate and Panelling will become familiar and clearly understood, allowing you to better understand the information being presented to you by the EPC.

You should also seek to understand dealership and manufacturer spare parts operations. How parts are warehoused, ordered, delivered and sold, understanding how a service technician works and is paid, the tools and information available to the service workshop, and the customer's view of the service and parts processes will all help you see where your work fits in to the entire operation, allowing you to make informed decisions when confronted with unusual situations.

Finally, you should continue to cultivate your desire to learn, use and share the expertise you gain from knowledge and experience. This is the one quality that, above all others, is the mark of the most successful Parts Advisors. With effort it is possible for anyone to achieve a high level of competence.

Enlightenment is available to any soul who is willing to seek it.

The best way to learn is to teach others.

4. The Mercedes-Benz Parts Numbering System

In this section, you will learn some important information about the Mercedes-Benz Parts Numbering System. It is essential that you understand the basics of the system and terminology, so you can order the correct part. Each Mercedes-Benz spare part is assigned an identification number, used for ordering the part. These part numbers follow an ordered system, as follows:

4.1. Basic Numbering

The most common part number has 10 digits and a letter prefix “A”. Such a part number is written as follows:

A 117 030 10 17

Note the spacing and grouping of the digits. When spoken, you say it as it’s written, like “A-one-seventeen, oh-thirty, ten, seventeen”. Most people used to MB part numbers will say it that way, and saying it differently tends to confuse.

Each section has a distinct meaning:

- **A** Prefix for Passenger Car parts. Other prefixes (such as B, C, N and W) are explained below.
- **117** Type Number – This is the internal type designation indicating which vehicle type for which this part was first designed. Parts may be (and often are) used in other types. Some parts use other numbers unrelated to model types.
- **030** The parts group number. Major group 03 corresponds to the EPC group to which the part generally belongs. The third digit indicates (broadly) the type of part from that group.
- **10** Modification Number. This identifies the modification status or variant of the part. The first version or variant is generally numbered 00, and numbers increase with new variants. A higher number does not necessarily mean the part is newer or better, it is assigned somewhat randomly and not all numbers are assigned or used in strict sequence.
- **17** Part Type. Most similar parts of a certain type have the same type number. In this case, part type 17 (in group 030) is a Piston.

4.2. Sorting order

When sorting a list of 10 digit part numbers, use the order shown:

A	117	030	10	17
	3	**1**	**4**	**2**

For example, first put all part numbers in order of the second group of digits (030), then sort within the group by type (17), then by Type number within Type (117), and last by Modification Number.

The following list is sorted properly:

A 117 030 10 17
A 117 030 12 17
A 001 350 29 45
A 001 350 01 46
A 001 350 11 70
A 003 350 09 72
A 210 545 21 86
A 124 545 01 99

All other part prefixes (such as N or B) are sorted in direct numerical order.

Examples of the spacing and grouping of part numbers with other prefxies are:

B6 782 0415
BQ682 0116
HWA202 545 27 19
N004019 013802
W210 589 03 29 00

4.3. Letter prefixes

Use of the correct letter prefix is essential. While most parts are numbered uniquely, some may be duplicated with different letter prefixes.

A Passenger car parts
B Accessory parts
BQ MBUSA Domestically-assigned part numbers
C Commercial vehicle parts
HWA Special passenger car parts (generally AMG)
N DIN (German Industry Standard) parts
Q Not a valid part, used as a placeholder in some EPC catalogs. Usually 12 digits, often all zeros. Do not confuse with BQ part numbers.
W Special tools
X Some commercial vehicle & light truck parts

4.4. Suffix Numbers

There are two kinds of suffix number, known as ES1 and ES2. A part generally uses only one or the other, but both may be used for a single number. ES is an abbreviation for the German words Erganzungs Schluessel, which translate as “Supplemental Key”

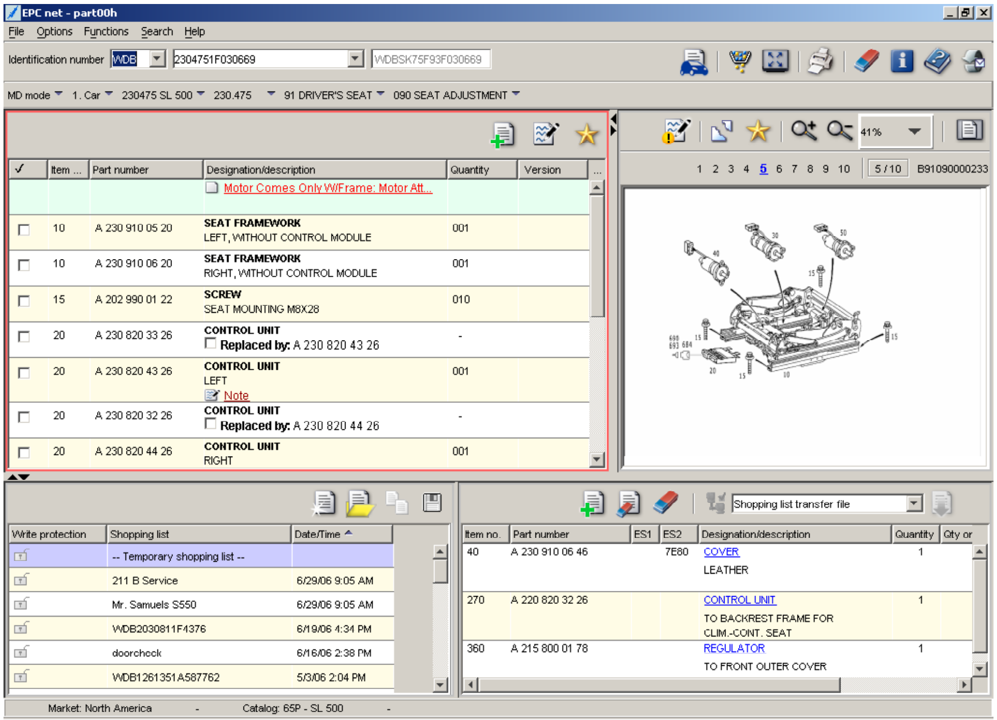
ES1 codes are always two digits, and always immediately follow the base part number. The ES1 is sometimes called an “Index” number in the literature. Some ES1 numbers can be used to designate a specific version of a base number, such as for Pistons or Springs, which can be supplied with slightly different sizes or values. The meaning of some common ES1 codes is as follows:

05 Special demand part. Often used to denote parts supplied for a service action
10-15 Package sizes for a base number. Often used for service fluids

26-28	Component that is pre-programmed or specially prepared
64-69	A part identical to the base number, but from a different supplier or location
70	A core with value which should be returned for credit
80	A remanufactured component, which also has a core value.
81	A remanufactured component, which does not have any core value.
87, 88	Same as 80
90	A remanufactured part with no core value, often used for Warranty repairs

ES2 codes are always four characters, with digits and letters often mixed. These are always placed at the end of the entire part number, after the ES1 number if necessary. These are most commonly used to denote the color and texture of specific parts which are supplied in different colors and/or textures. The correct ES2 code must be specified to receive the correct part. These are sometimes called “Color Codes”, not to be confused with paint and trim color codes. Some parts have an ES2 code of 9999, which generally means that the part is provided in primer and must be painted.

5. Features and Functions in EPC net

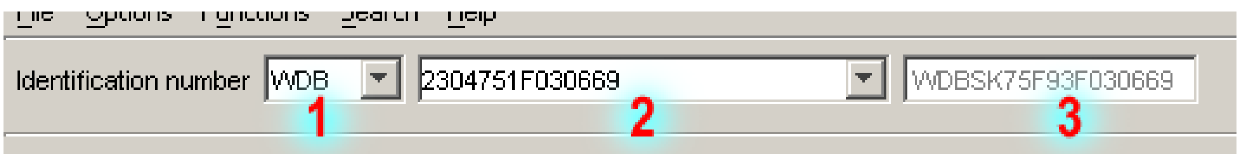


5.1. Toolbar & Icons

The toolbar has two sections: The VIN fields and the Function Icons.

5.1.1. VIN Fields:

There are three text boxes that are used to specify the VIN of the vehicle for which you are specifying parts:



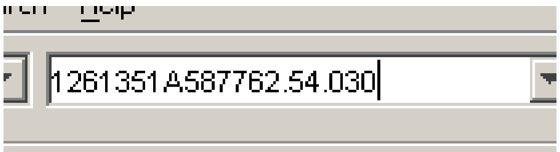
· **Manufacturer Plant Code (1):** Use this box to specify the first three characters of the VIN, which is known as the Manufacturer Plant Code. Most vehicles assembled in Germany use the plant code WDB, although newer vehicles use WDC, WDD or others. Vehicles assembled in Vance, Alabama use the plant code 4JG. To change the Plant Code selection, click on the small arrow to open the list. Alternatively, simply type in the plant code at the beginning of the FIN or VIN in box 2 as explained below.

· **VIN or FIN (2):** The VIN is the Vehicle Identification Number specified for USA vehicles as of Model Year 1980, having a form like CB35DXMA123456. Note that the plant code may be omitted when entering data into this box only if it is correctly selected in box 1, otherwise you should enter it, as in WDBCB35DXMA123456. Doing it this way saves time when copying a VIN or FIN from another source, such as Dealer Management System (DMS). The FIN, also known as the World VIN or Chassis Number, is the baumuster, version, plant identifier and sequential six-digit chassis number, such as 1261351A123456, again with or without plant code.

Vehicles before Model Year 1980 have only a FIN. Please refer to the section on VINs, FINs and Baumusters for further information.

The VIN field has a History feature: To bring up a VIN or FIN previously entered, click on the small arrow to open the list. The list contains approximately the last 50 entries, and can be cleared by accessing the My Profile / Advanced Preferences / History screen from the initial sign-on screen.

The VIN field also supports a shortcut feature, which allows navigation directly to a group and subgroup by specifying it after the VIN, separated by periods. For example, typing in 1261351A587762.**54.030** opens Group 54, subgroup 030 for that VIN. You can select only a group without a subgroup if desired.



· **The display field (3)**, into which you cannot enter data, changes depending upon where in the EPC you have navigated. In the main parts window, the VIN is displayed (Box 2 changes to the FIN). In an Aggregate catalog, the serial number of the appropriate Aggregate (such as Engine) is displayed

5.1.2. Function Icons

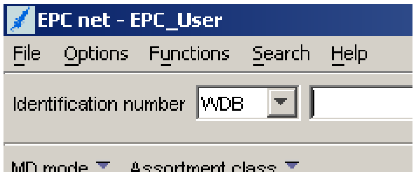
There are nine function icons displayed on the Toolbar:

- **(1) Datacard** – See Section 5.8.1
- **(2) Creat/Show Note for Group or Subgroup** – See Sections 5.8.3 and 5.8.4
- **(3) Filters** – See Section 5.2.2
- **(4) Maximize** – Removes or adds the Shopping List and Shopping List Manager view form the screen.
- **(5) Print** – See Section 5.2.1
- **(6) Erase** – Deletes the selected data
- **(7) Information** - This is the same as selecting Help About from the Menu Bar
- **(8) Help** – Opens the context-sensitive Help feature. See Section 5.2.5
- **(9) Instant Feedback** – Opens a window allowing an easy way to report EPC errors. Depending upon your EPC server's configuration, this will open either a FAX Form window, which can be printed and faxed to MBUSA Dealer Parts Services, or an E-Mail Form window, which allows the form to be e-mailed to MBUSA. Be sure to include your name and phone number! Contact your EPC service provider is this feature is not activated on your server.

Note that this feedback form is not for technical assistance, it is only for reporting errors in the EPC data or application. Be sure to be viewing the correct part or illustration when you click the icon, otherwise the incorrect information will be gathered for the report. **Dealership personnel are strongly encouraged to report all errors**, and every reporter will be contacted within a few days to acknowledge the report. For technical issues with the EPC in your dealership, contact ProQuest in the USA and Canada, or your service provider in other countries.

5.2. Menus

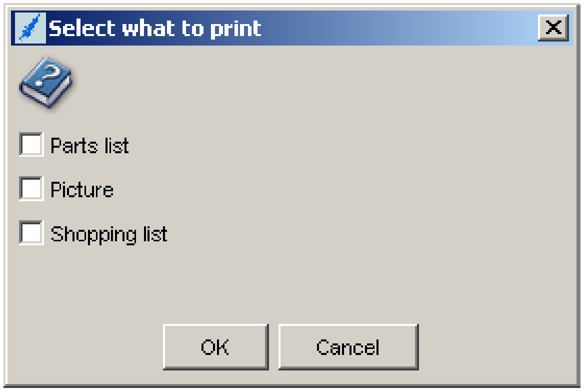
There are five menu choices:



5.2.1. File

The File menu has two choices: Print (Ctl+P) and Exit (Alt+F4).

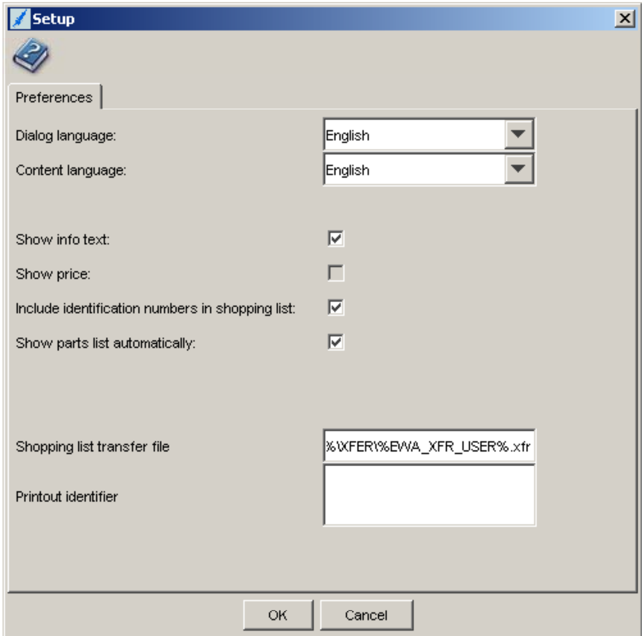
· Print opens a dialog box allowing you to print the parts illustration, the parts window contents, and the Shopping List. Check off the item(s) you want to print and click OK. A Windows Print dialog box will open for each item selected. This has the same function as clicking the Print icon.



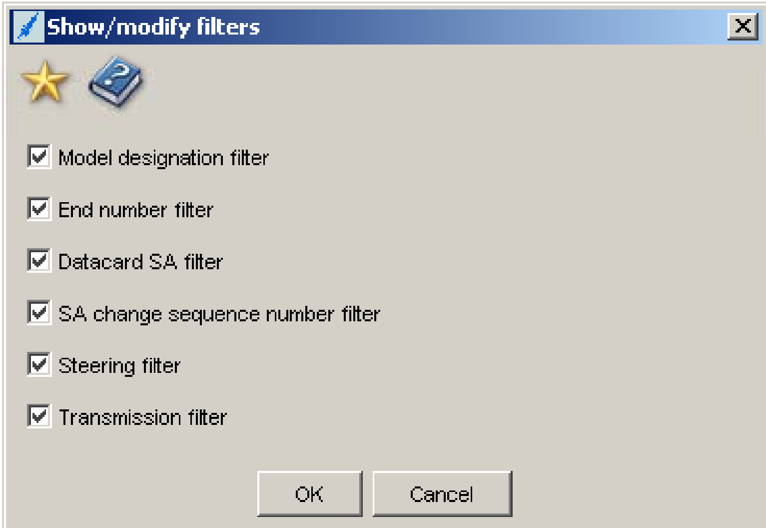
· Exit closes the EPC net program. Remember to also log off from the server after closing EPC net.

5.2.2. Options

The Options menu has two choices: Setup (F2) and Filter (F7)



- **Setup** opens the preferences pane, allowing you to select:
  - The language of the EPC net application (such as menus and help) and, independently, the language for the parts data. These can be changed at any time, for example in the middle of looking up a part, as desired. Choices include English, French, German, Italian, Spanish, Portuguese, Polish, Japanese Chinese and Turkish. The data is not available in Polish or Turkish.
  - Checkboxes for optional Info Text (displays special information messages regarding the specific vehicle when the VIN is entered), Price (which displays a price associated with parts in the Shopping List), VIN shown in Shopping List (the VIN is displayed in the shopping list) and Show Parts List Automatically (can speed performance when unchecked by not showing unselected parts). It is strongly recommended that the Info Text box is checked. Note that the Price functionality is not enabled by MBUSA at the time this was written.
  - The path and name of the Parts Transfer File, which is delimited ASCII text. If you click on the “Export to DMS with selected session” button and the session “Shopping list transfer file” is selected, the Shopping List will be placed into a file with the location and name as specified here. If your dealership EPC net system has Dealer Management System (DMS) Data Transfer enabled, this file becomes less important, since you will usually transfer the Shopping List to a Repair Order in your DMS.
  - Printout Identifier, which is text you can enter to appear on each printed page, useful when a printer is shared by many users.



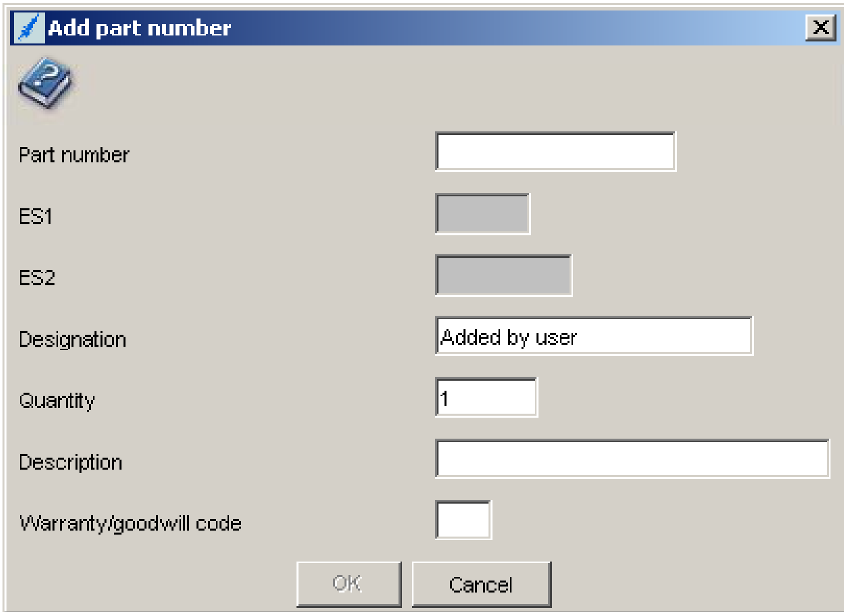
· **Filter** opens the Show/Modify Filters dialog box, allowing you to review or set the current parts filter level. This has the same function as the Filter icon. You can only deselect filters from the current default level. For example, if you enter only a Baumuster and not a full VIN, it is not possible to select End Number (VIN) filtering. In most cases, the default filters will provide the best performance, however there may be cases where removing one or more filters will allow a view of additional parts, which may be helpful in selecting the correct part.

- Model Designation Filter: Filters out parts inapplicable to the Baumuster selected. With this filter off, you can see to which Baumusters each part applies.
- End Number Filter: Filters out parts that do not apply to the specific VIN end number, based on footnotes. In cases where the VIN end number is close to the footnote end number, it may be more accurate to disable this filter and use the production date instead. In any case, vehicles close to the cutoffs in the footnotes should be inspected to ensure the correct part is selected.
- Datacard SA Filter: Filters out parts that do not apply based on datacard information (Codes, SAs). Disable this filter when it is known that the vehicle data card does not have the correct code or SA marked (e.g., Code 264 "License Plate Attachment Americas", which may be missing from late 1990's and early 2000's cars)
- SA Change Sequence Number: SAs used in older cars may have different versions, designated by a two-digit Change Sequence number. Disabling this filter shows all SAs with the same 5-digit base number.
- Steering Filter: This filters out parts for cars which are right hand drive when looking at LHD vehicles, and vice-versa. These parts have an "L" or "R" in the Version column of the Parts Window.
- Transmission Filter: This filters out parts for transmissions (manual/automatic) that do not apply to the specific vehicle, but may be in other vehicles of the same body type.

5.2.3. Functions

The Functions menu has four choices: Add Part Number (Ctl+F9), Datacard (Ctl+F6), Footnote (Ctl+F2) and Note (Shift+F2).

- Add Part Number opens the Add Part Number dialog box, allowing you to manually add any part number to the Shopping List.

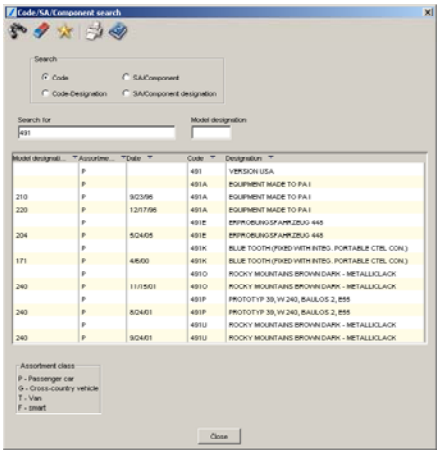


- Datacard opens the datacard display window, if a valid VIN or FIN has been entered. The vehicle data card contains all the important information about the vehicle equipment and configuration, which is often necessary to specify spare parts correctly. This has the same function as the Datacard icon. Refer to Section 5.8.1 for more information about the Datacard
- Footnote opens the Footnote window, displaying all footnotes applicable to the selected part, and allowing for a search of footnotes within the selected parts group. This has the same effect as clicking a footnote hyperlink in the Parts Window. Refer to Section 5.8.2 for more information on the Footnote window.
- Note opens the Add User Note dialog box. Please refer to the section “User Notes” in section 5.8.3 for a complete description.

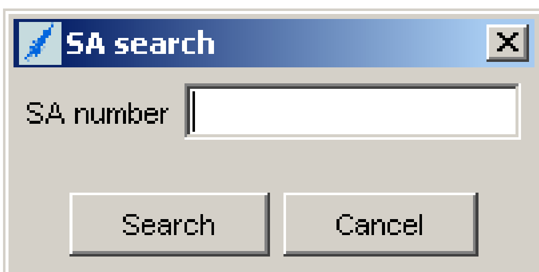
5.2.4. Search

The Search menu has three choices: Code/SA/Component (Shift+F8), SA (F10), and Part (F11).

· Code/SA/Component opens the Code/SA/Component Search dialog box. Here you can search for the meaning of a specific code (the three or four character equipment, paint or trim code), or a specific code designation (the text describing a code). SA/Component and SA/Component Designation searches are only valid for commercial Vehicle applications, and do not function for Passenger cars. To search Passenger Car SAs, use the next menu item.



· SA opens the SA Search dialog box. An SA is an abbreviation for the German term “Sonder Ausstattung” which translates as Special Equipment, or Options. This should not be confused with three or four character equipment codes. SA numbers are shown on the vehicle data card, but have not been used for Passenger Cars since the mid-1990s. In older vehicles, especially Classics, it is important to look in the correct SA Catalog to identify the correct parts.



· Part opens the Part Search dialog box, which has three tabs. Please refer to Section 5.9 for a complete description of this powerful tool

- o Part without vehicle data.
- o Part with vehicle data
- o Vehicle model designation for major assemblies

5.2.5. Help

The Help menu has four choices: EPC net Help (F1), What’s New (Shift+F1), and About EPC.

· EPC net Help is one way of entering into the extensive and detailed on-line help system. If you are in a specific window of the EPC, the context-sensitive feature opens the help information most relevant to that window. You can always go to the help index and search features. Use this whenever you have a question about how something works.

· What’s New: This section of EPC net provides a monthly update on major changes and new functionality in the EPC. It is a good practice to review this information after each data update, about once each month.

· About EPC: Displays copyright notices, version information for the EPC net software and data, as well as the default languages selected. When seeking support or EPC assistance, it is useful to know the EPC application and data versions.

5.3. Index bar

There are a number of Pull-down menus that appear on the Index Bar, which are used to navigate through the EPC and then show to where in the EPC you have navigated.

1. Mode box – Select MD mode to work in Model Designation mode (this is the usual mode). SA mode opens the SA catalogs, and allows lookup by SA. Paint/Operating Fluid mode opens the Paint/Operating Fluid catalog, allowing you to look up the part numbers for paints, service fluids, and materials.

2. Assortment Class box – Select the type of vehicle from this box. In most cases, you would choose Car. Note that the M Class and GL Class for USA are considered Cars, and the G Class (such as Model 463) a Cross-Country Vehicle. Depending upon your EPC net License level, you may see other vehicle types here.

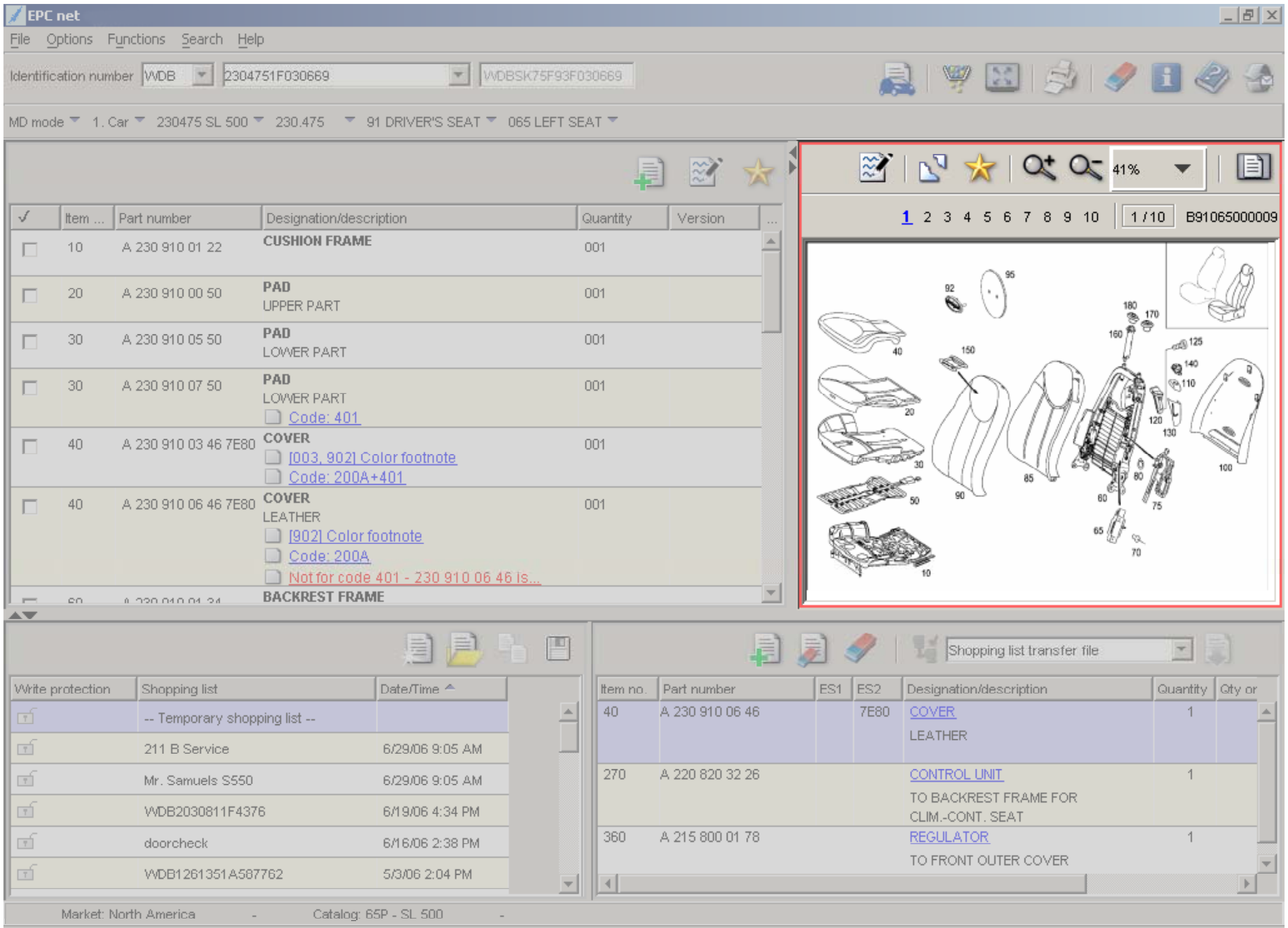
3. Model Designation box – select the model designation (either Baumuster, aggregate or Model name) here. Typing into the text box has a filtering action, removing all entries that do not contain the typed characters. Note that the best practice is to type in the complete VIN in the VIN field, see Section 5.1.1.

4. Group box – Select the parts Group or Aggregate from this menu.

5. Subgroup box – Select the Subgroup for the selected Group from this menu. Also displayed are applicable SAs and Market Notes, see Section 5.8.4.

6. Special boxes – these appear when necessary to display a finer gradation than possible with the Subgroup menu. Generally self-explanatory.

5.4. *Illustration Window*



The illustration window shows the parts illustrations, used as a guide to select the part callouts. Selecting a callout by clicking on it in the illustration causes the Parts Window list to show only those callout items selected. Multiple callouts can be selected.

Also see Section 6.8 for details about working with the Illustrations.

At the top of this window are some icons:

· The **User Note** icon allows the user to display or add a User Note to the illustration. A yellow frame alerts the user to the presence of a User Note (see Section 5.8.3), while a res frame indicates a Market Note (see Section 5.8.4).

· The **Tear-Off** icon allows the illustration to display in a separate full-screen window. This is especially helpful to examine fine details on an illustration.

· The **Star** (select all) icon deselects all callouts

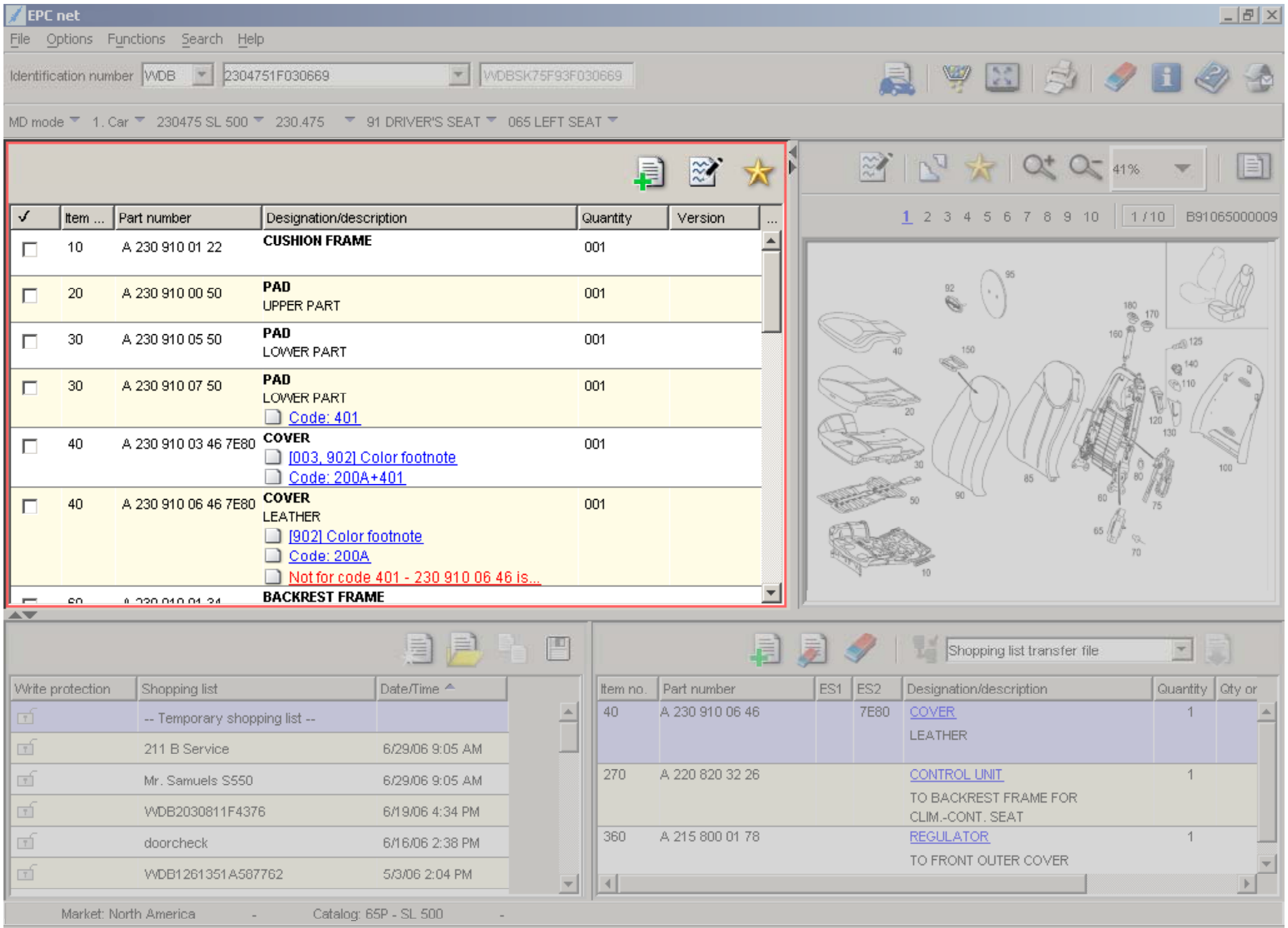
· The **Plus (+)** and **Minus (-) Magnifying Glass** icons cause the illustration to zoom in or out, respectively, but does not change the overall size of the Illustration window. Navigation within a zoomed window is done by clicking and dragging the mouse cursor, or with the scroll bars that appear.

· The **Zoom Menu** is used to select a specific zoom level, similar to the magnifying Glass icons. It also displays the current zoom level.

· The **Fit To Window** icon causes the illustration to zoom so that it fits inside the window. You can also drag the bars (See Section 5.8.5) to change the dimensions of the window.

Beneath the icons are the **Picture Navigation Bar** and **Illustration Number**. The Picture navigation bar is used to navigate to all of the illustrations within the group, including across subgroups. Blue numbers are hyperlinks; simply click on them to switch illustrations. The keyboard left and right arrow keys can also be used to jump to an adjacent picture. When more than ten illustrations are available, arrows at each end become active; clicking on them brings you to the next set of ten illustrations. A box shows which illustration you are viewing, and the total number of illustrations in the group. The Illustration Number identifies the specific illustration you are viewing.

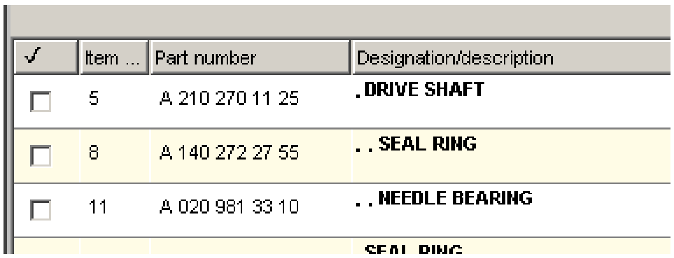
5.5. *Parts Window*



The Parts Window lists the part information corresponding to the illustrated parts shown in the Parts Illustration Window. Note that not all parts are necessarily illustrated – in these cases, the part number will be listed without a callout number - and that not all illustrated parts are listed - in these cases, the parts do not apply to the vehicle and are filtered out. Clicking on a part line will cause it to have a blue background, and the callout associated will be highlighted in the illustration. Hyperlinks may be visible for Footnotes, Codes, User Notes and Market Notes. Details for working with these are given later in this document.

To add a listed part number to the Shopping List, simply click on the checkbox to the left to add a check mark. The column sizes can be adjusted by clicking and dragging the lines between the column headers. Be sure to pay attention to the Quantity and Version columns; although they may be hidden, they can contain important information.

Some parts are subassemblies of larger parts, and this is indicated by one or more preceding dots before the part name. Item 5 has a single dot in front of it, because it is part of a larger assembly (the automatic transmission in this case). Items 8 and 11 have two dots, indicating these two items are components of Item 5, but also available separately. While this convention holds true in nearly all cases, parts availability may cause changes in which parts are included with a larger assembly – this is more common in older cars.

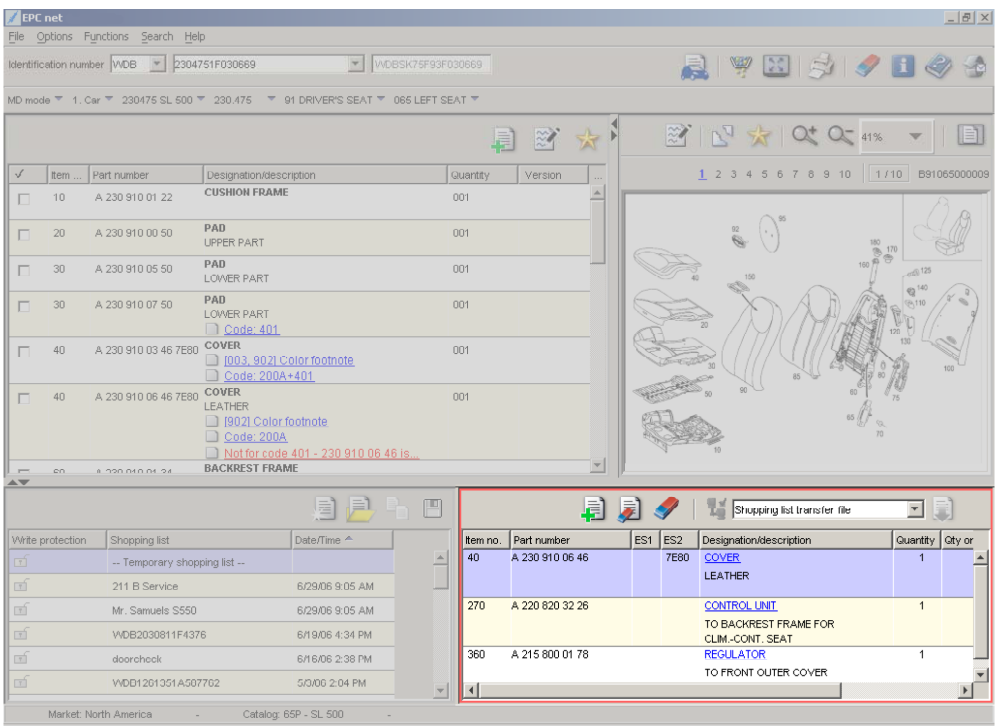


✓	Item ...	Part number	Designation/description
☐	5	A 210 270 11 25	. DRIVE SHAFT
☐	8	A 140 272 27 55	. . SEAL RING
☐	11	A 020 981 33 10	. . NEEDLE BEARING

At the top of this window (see illustration on previous page) are some icons:

- The **Add Part Number** icon allows you to add a number that is not listed to the Shopping List. Examples of such numbers are accessory, tool and operating fluid numbers.
- The **Create / View User Note** icon is used to create, view or manage a User Note attached to a specific part. See Section 5.8.3.
- The **Star** (select all) icon causes all parts to be displayed in the Parts Window.

5.6. *Shopping List Window*



The Shopping List window is a place to store a list of part numbers that you have specified until you want to move on to the next step – either ordering the parts or assigning them to a specific Repair Order or Quote. The shopping list works in conjunction with your Dealer Management System (DMS) and the Shopping List Manager, described below.

5.6.1 Icons in the Shopping List

· To place a part number on to the Shopping List, simply click the check box next to the part number at the left side of the Parts Window. Unchecking the box will remove the part from the list, as will the Remove Selected Item icon.

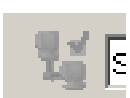
· You can clear the entire shopping list by clicking the Delete All icon.

· You can also add a part number not listed in the EPC, such as a BQ or Accessory number, to the shopping list by clicking the Add Part Number icon

The most valuable property of the Shopping List is that it is **Context-Based**. This means that the part numbers on the shopping list are associated with Hyperlinks. Clicking on the hyperlink will not only bring up the correct Parts and illustration Windows, but will also re-populate the correct VIN under which the part was originally put onto the list. This time-saving feature has many uses, such as simplifying a return to a previous job when a lookup is interrupted.

To transfer the contents of the Shopping List to a Repair Order or Quote in your DMS, simply click on the Export to DMS icon.

The exact function of this icon depends on the selection in the text box above the Shopping list, and the specific DMS emulation installed. Although the DMS Export feature should be pre-configured, settings may be changed by clicking on the Integration Configuration icon shown below.

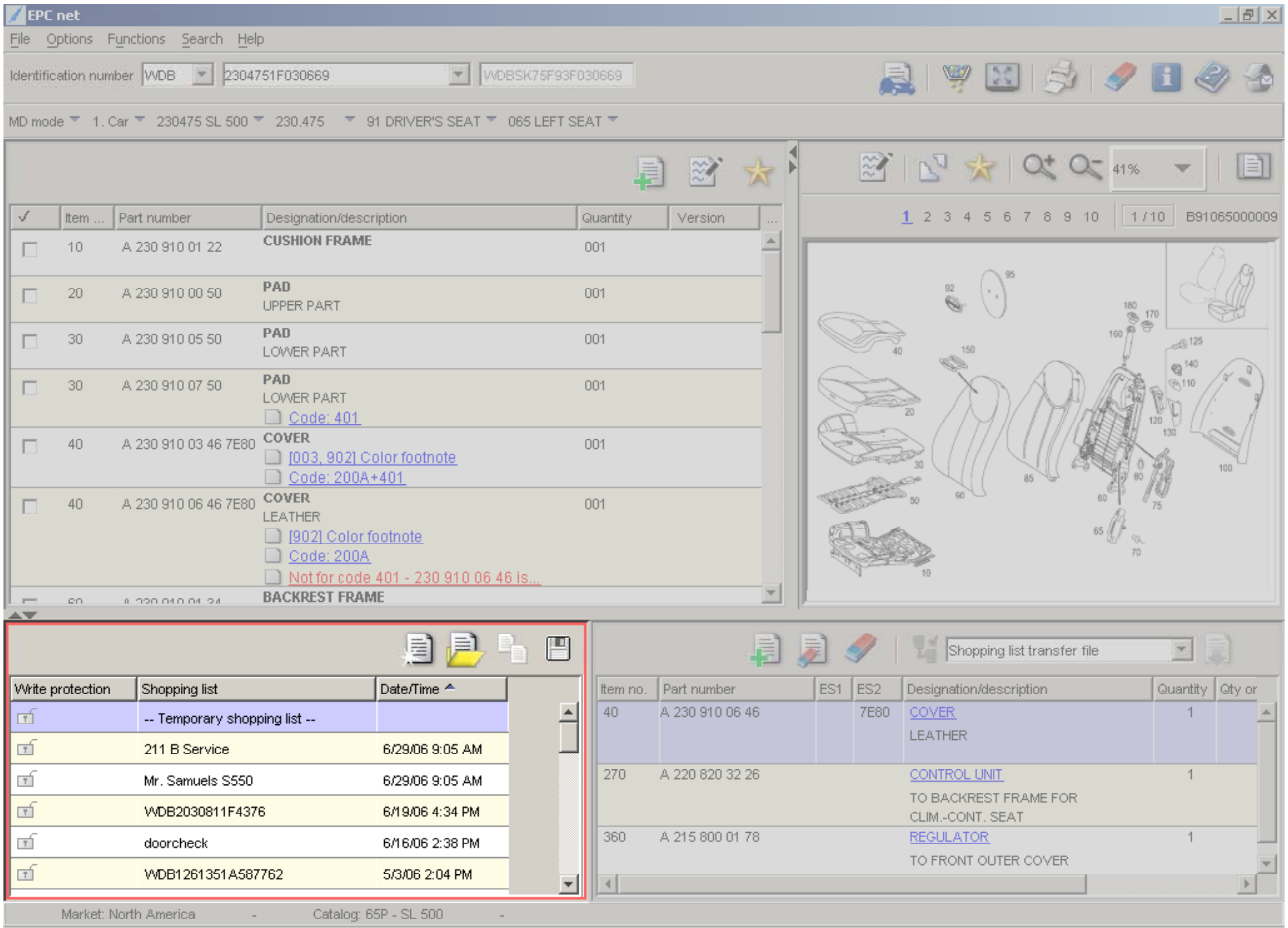


The basic configuration (“Shopping list transfer file”) allows shopping lists to be exported to flat text files with comma separated values. The name and location of the file is specified in the Setup menu.

5.6.2 Modifying the Shopping List

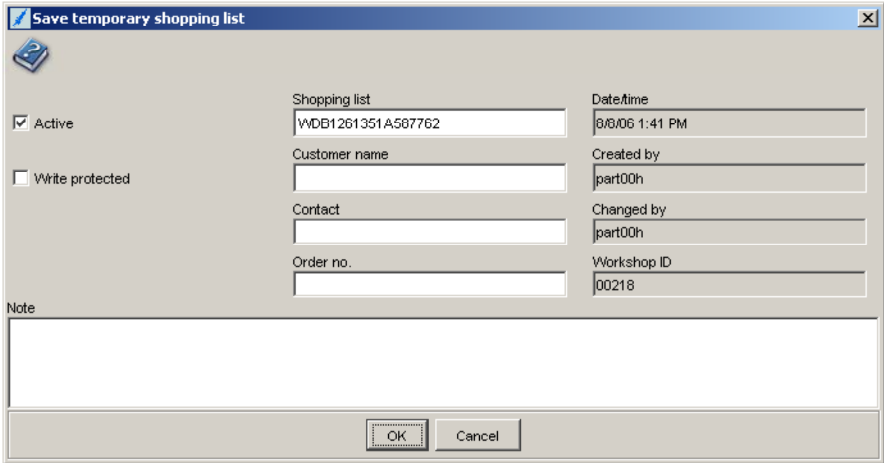
You can add to, delete or change the information shown in the fields ES1, ES2, Description and Quantity by simply clicking in the field and typing in the changes. This can be especially helpful for the ES1 and Quantity columns, and for adding more information to the Description column, beneath the hyperlink. (If you click on the hyperlink, you will navigate to the part context instead).

5.7. Shopping List Manager



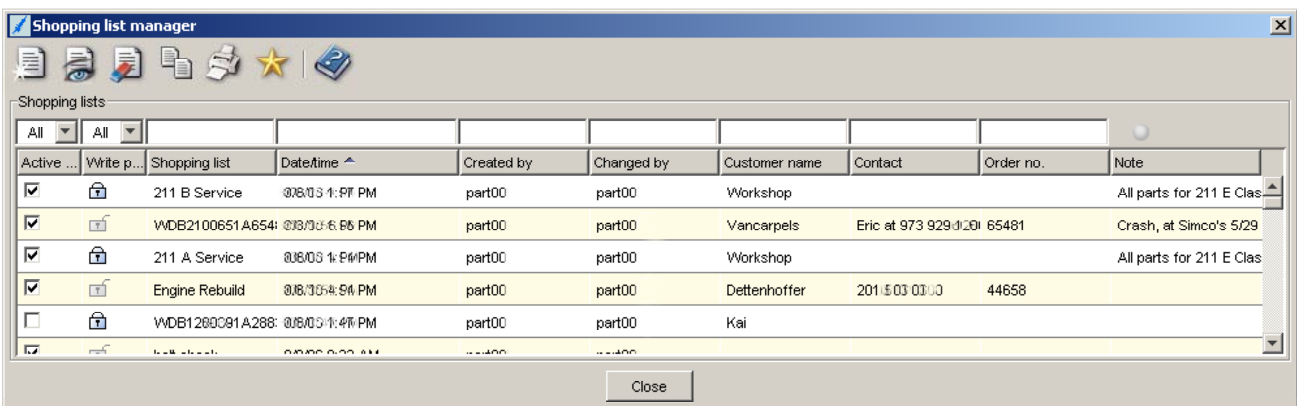
The Shopping List Manager is a powerful tool, allowing you to manage and better use Shopping Lists to save time and effort. The four basic icons allow you to create a New List, enter the List Manager feature, Copy a list, or Save a list

Creating a new Shopping List is as easy as saving the current Shopping List, which until saved is called the Temporary Shopping List. Alternatively, click on the New List icon. Clicking the Save button brings up a menu where you can enter information about the list, which may be helpful when later reviewing the list. The default name of the list is the VIN, but if a list with the same name already exists, you need to choose a different name. You will be prompted to save or ignore any changes to a list.



In the shopping list save window, you can specify the customer name, contact information, repair order or other number, free-text notes about the list, as well as setting the list to active or not, and write-protected or not. An Active list is visible in the Shopping List window, and a Write-Protected list cannot be changed unless the write protection is removed.

Note that the temporary shopping list is only available on the specific computer on which it was created, while saved lists are available to everyone using the same EPCnet server, which is usually the entire dealership. In a standalone installation, as well as an online situation, saved lists are visible to only the user who created them.



The Shopping List Manager window allows you to manage your saved lists. The icons along the top of the window are to:

- · Create a new list
- · Open a single selected (highlighted) list
- · Delete all selected lists (Write-protected lists cannot be deleted)
- · Copy the selected list
- · Print the selected lists
- · Remove all filters
- · Show Help

A useful feature is the ability to filter your saved lists by any criteria. At the top of each column are text boxes; the first two have limited selections, while the remainder accept any text. The display of lists will be reduced according to the text entered. For example, if the text "211" is entered into the Shopping List name field, only those lists with “211” somewhere in their name will be displayed. As another example, to find a specific customer’s shopping list simply start typing their name until the list filters down to the point where the desired list becomes visible. This assumes, of course, that the customer’s name is used somewhere in the list.

This brings up an important point: You or your Management needs to develop a consistent way of naming and storing shopping lists, if you are to gain the full value of this tool.

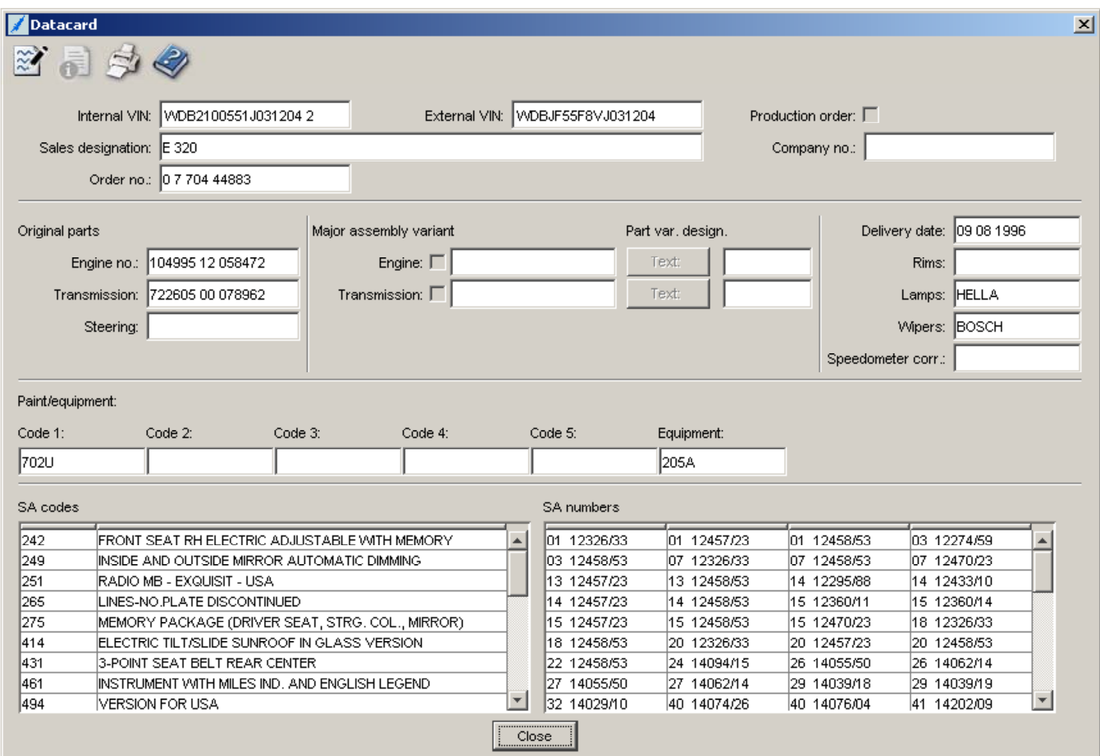
The idea is to create shopping lists for each typical service job by model and, if necessary, model year. All of the A and B services, along with common jobs like brakes or an accessory installation. This will save time later, instead of looking up each part for the job, the saved list can be called up and, if necessary, copied and changed before exporting it to the Repair Order.

By using a consistent system, these lists – some dealers have over a hundred – can be quickly and easily found and used. If an inconsistent naming system causes you to use more time finding the list than it would take to create a new one, then no time is being saved.

Be sure to write-protect any common lists that are created, to ensure they are not changed or deleted by mistake. Also, keep only active lists visible, to reduce clutter on the list manager display. With some thought and effort, significant gains in efficiency can be realized.

5.8. Other Features & Functions

5.8.1. Data Card



The Vehicle Data Card display shows all relevant information concerning a specific vehicle as it left the production line. Note that in most cases any retrofitted systems or accessories are not displayed, nor are the new serial numbers of an engine or transmission that was replaced.

With the availability of the FDOK-Web system in the United States, the data card information is also visible. It should be noted that the FDOK system is the source of the EPC data cards, as well as the vehicle information in Vehicle Master Inquiry (VMI). FDOK-Web may also contain additional information about the vehicle, and is especially useful for vehicles built too recently for the data card to be included in the EPC.

Looking at the sample data card, we see the following information:

Internal VIN (the “FIN”) and USA VIN
Model sales designation
The original production order number
The “delivery date”, the date it left the production line, in DD MM YYYY format
The exterior paint color (Code 1)
The second body color (Code 5) if applicable
The interior color (Equipment)
A list of the Optional Equipment (SA Codes) installed
A list of any SA numbers (additional optional equipment, see section 5.2.4)

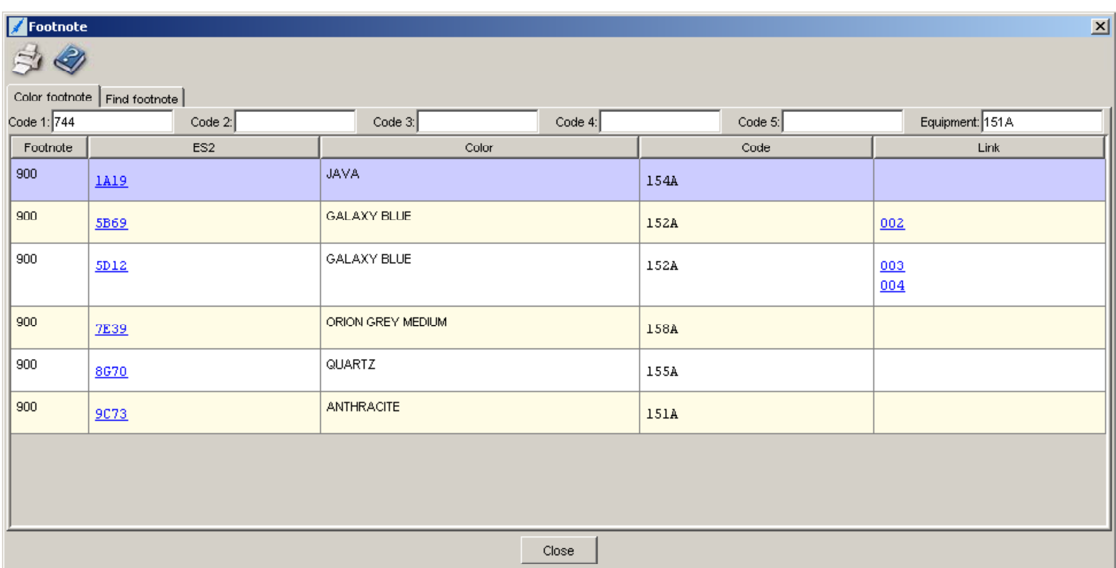
Not all information is shown for every vehicle. Older vehicles generally have a long list of SA Numbers, while newer vehicles do not use them. The production date may be missing or obviously false on older vehicles. Some Option Codes are ‘standard’ and may not be shown on the data card; it is assumed that the user knows of these, but in many cases that may not be true.

Notes about or missing information from a particular Data Card can be added by clicking on the “add User Note” icon at the upper left of the window. See sections 5.8.3 and 5.8.4 for more information about this feature

The data card is an essential record which is required to properly specify most parts. It may prove helpful to print the data card for reference while performing difficult specifications.

5.8.2. Footnote Window

Footnotes are one of the most important pieces of information provided by the EPC to help select the correct part, and must be understood clearly.



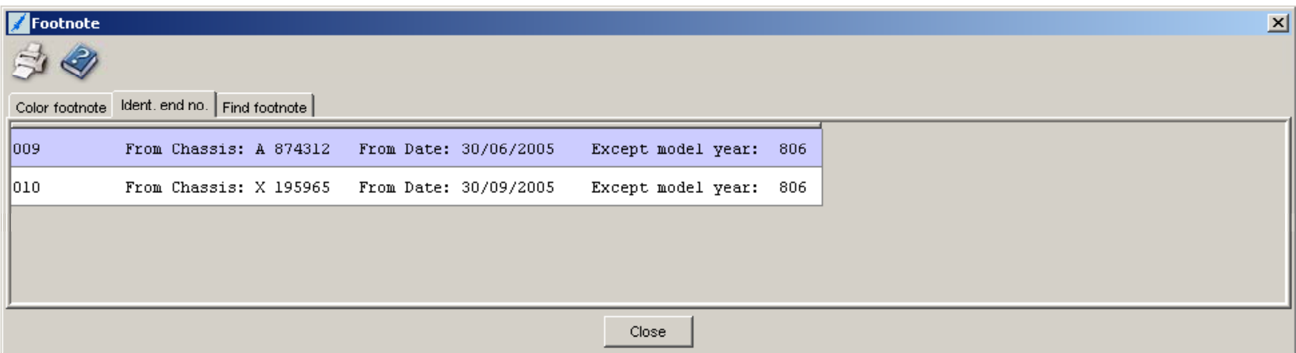
Footnote	ES2	Color	Code	Link
900	1A19	JAVA	154A	
900	5B69	GALAXY BLUE	152A	002
900	5D12	GALAXY BLUE	152A	003 004
900	7E39	ORION GREY MEDIUM	158A	
900	8G70	QUARTZ	155A	
900	9C73	ANTHRACITE	151A	

In the image on the previous page a typical color code footnote screen is shown. On the left is the footnote number: Be sure to look at the correct footnote number specified under the part number (900 in this example), since multiple footnotes may be shown on the same screen.

Towards the right, the trim color codes (154A, 152A, etc) are seen. At the upper right, in the box labeled “Equipment” is the vehicle’s trim color code, taken from the vehicle data card, which is 151A in this example. (The number 744 at the upper left is the exterior paint code). Looking in the Code column, the last entry also has trim color code 151A, and so the correct ES2 for this part is 9C73, Anthracite. If the trim color code is not listed, the part selected does not apply to the vehicle.

Also notice that for some of the lines, there is a hyperlink in the “Link” column: These links must also be reviewed (for trim 152A in this example) to further determine the correct ES2 code. A common source of errors is a failure to review these links.

The footnote window often has multiple tabs at the top; the one in the example has only two. It is very important to review the information on all of the tabs, since they contain information that is also relevant to selecting the correct part.



The image above shows the type of information that is often seen on the “Ident end no.” tab. Without this information, the wrong part may be specified. Refer to Section 6.6 for more information on footnotes.

5.8.3. User Notes

A powerful feature in EPCnet is the ability for the user to write notes or comments and attach them to a Part Number, Illustration, Group, Subgroup or Data Card. When you attach a note to an illustration, you can choose whether the note appears in the one Catalog, or wherever that illustration is used in all Catalogs. When you attach a note to a part number, you can choose whether the note appears in just that one case, in all instances of that part number in a single Catalog, or in all instances of that part number in all Catalogs.

To create a User Note for a Part Number, first select the Part Number by clicking near it (the background turns blue) and then select the Write Note icon located in the Parts List. Enter the desired text in the bottom text box, then decide if the Note should be shown “Only in this catalog (Model, Group, Subgroup)”, “Only in this catalog (in all groups)” or “In all standard catalogs with this part number”. Finally, click “OK” to save the Note. To create User Notes for other items, simply click on the Write Note icon in the area of that item.

Write Note Icon

User Notes appear beneath the **Part Number** as a Note symbol with a yellow border and a Burgundy hyperlink “Note” (far left), for a **Group or Subgroup** as the yellow-border symbol next to the Group or Subgroup line (left center), for an **Illustration** as a larger symbol also with a yellow border (center right) and for a **Data Card** as a yellow border surrounding the Data Card icon (far right). To see the note, simply click on the “Note” link, or the icon for the other cases.

User Notes can be seen by all users connected to the same server. In the case of a standalone installation, or in online use, only the specific user can see the note, they cannot be shared.

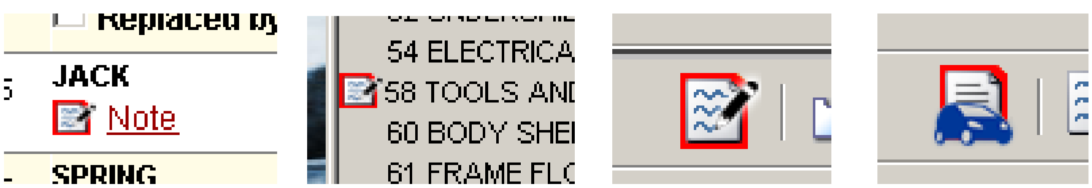
User notes are an excellent way to alert others on your team of special conditions concerning a part number, or to assist with specifying the correct part. For example, you might write a note explaining that the technician needs to specially prepare the part for warranty return, or that an EPC error was found. (Please help your fellow specifiers by reporting all EPC errors, so we can create a Market note! See Section 5.1.2, item 8).

User Notes are managed on the Administration tab. There you will find tools to view, delete, change and print all User Notes. Much like the Shopping List Manager, you can also filter the list to display specific notes.

5.8.4. Market Notes

Market Notes are similar to User Notes, but these are written and managed at the Manufacturer level, such as MBUSA. These notes, which cannot be edited or deleted by users, are inserted to bring attention to information that may be of value to someone specifying parts. They appear in the same ways as User Notes, however with a red border instead of yellow.

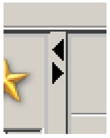
As with User Notes, the presence of a Market note is indicated in various ways, as shown below.



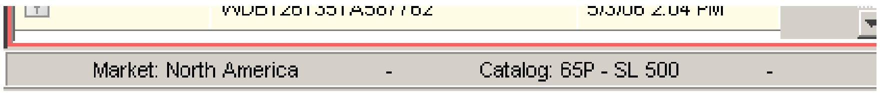
It should be common practice to review any Market Notes you encounter while specifying. In many cases, these notes contain exactly the information you need, and can save you considerable effort. Many of these notes are derived from Feedback reports received from dealers. Since the timeframe for a correction to appear in the EPC may be a few weeks, Market Notes are used to ensure the information is available within a few hours.

Dealers using a Local server at their dealership (as opposed to the Online systemn accessed through NetStar or the Internet) need to update their Market Notes file regularly, since this is not done automatically. The PAC website contains information on this issue.

5.8.5. Other Window Navigation



The bars between each window can be moved by clicking and dragging, as indicated by the arrows. For the vertical bar between the Parts and Illustration Windows a double-click selects between vertical divider (default) and horizontal divider. For the Horizontal bar beneath the Parts and Illustration Windows, a double-click maximizes these windows, removing the Shopping List and Shopping List Manager windows from view.



At the bottom of the window in maximized view (MS Windows functionality) the relevant Market and Catalog information is shown.

5.9. Parts Search

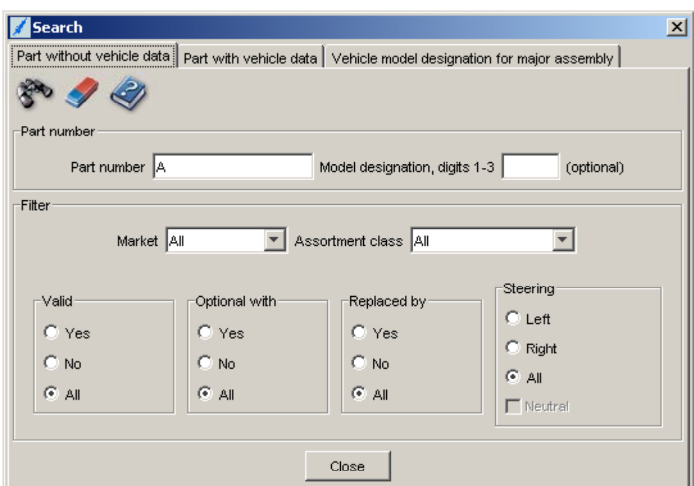
The Parts Search feature, accessed either through the Search Menu or by pressing F11, has three basic search types:

· Part without vehicle data

· Part with vehicle data

· Vehicle model designation for major assembly

5.9.1. Part without vehicle data



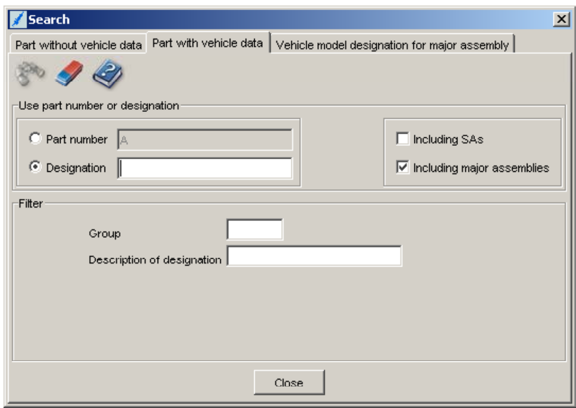
This search is used to find where a specific part number is used, across all standard catalogs. Simply type in the part number and click the Search icon (binoculars) at the upper left.

When searching, the letter prefix must also be specified, in addition to the part number. Do not include any ES1 or ES2 codes. Only a complete part number may be entered, there are no wildcard characters or partial entries allowed. The default letter prefix is “A”, but this can be changed if necessary.

The scope of the search can be limited in three ways: By 3-digit chassis or aggregate type (e.g., 211 or 722), by Market (e.g., North America), and by Vehicle type (e.g., Car, Bus, etc). Filtering can also be extended by whether the part is currently valid, is optional with some other part (i.e., either part can be used), is replaced by some other part in the EPC, or is for LH or RH steering only.

In general, most searches are performed with only the North America filter set.

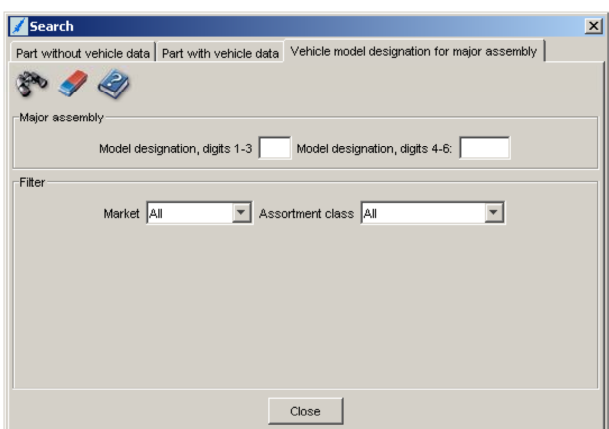
5.9.2. Part with vehicle data



This search is used within a specific vehicle record (i.e., by VIN) to search within a specific Catalog. You can search by part number (as above) or by the Designation (name) of the part, and filters for the group and the additional description text (the secondary text, such as ‘X27/3’ or ‘Left’). The search can also be expanded to include SAs, but note that newer cars do not use SAs.

Searching on the designation can be very useful. As an example, to find all parts with the designation “Relay”, simply type that text into the search box.

5.9.3. Vehicle model designation for major assembly



This search is used to find which vehicles use a particular aggregate (engine, transmission, etc). Simply enter the main type (such as 722 for an automatic transmission) and the sub-type (such as 901) to find all vehicles that use transmission 722.901.

As before, the search can be limited by Market and Assortment Class.

6. Basic Parts Specifying

In this section, a simple step-by-step process for identifying the correct part will be presented. In most cases, following this process will lead you directly to the correct part. There will be, however, some cases where further research is required before the correct part can be identified – this is known as Advanced Specifying. It is important to master Basic Specifying before you can gain skill in Advanced Specifying.

6.1. A simple process

The first step to any parts specification is to understand exactly what it is you are looking for. If you don't understand what the part is, what it does, and where it fits into the car, it is unlikely that you will recognize the correct part when you see it.

It can be helpful to have the workshop technician with you when you search for the part. They possibly know all about the part, and can help you. They have a vested interest in getting the right part, since it is their time lost when the wrong part is presented. The technician can also help you understand the part and its function, since they are likely working with it in the vehicle they are servicing.

The importance of this first step cannot be overstated: Without a full understanding, your search will be difficult, confusing and frustrating.

In addition to gathering information about the part, you need to gather information about the vehicle into which this part is to be installed. Nearly everything you need to know about the vehicle is contained in the Vehicle Data Card, which can be found in the EPC using the Vehicle Identification Number. The VIN is usually found on the workshop Repair Order, but you may need to ask the technician for it, or look at the vehicle in person. Since the early 1980's, the VIN is visible through the windshield, on the driver's side. In older vehicles, consult the owner's manual for the location of the VIN, but often this is located on a tag on the vehicle's "B" pillar.

Also note that in vehicles built before around 1982, the data card is not available in the EPC. For these vehicles, there is usually a Vehicle Data Card with the owner's manual and other vehicle papers. If it is missing, contact the MB Classic Center for a copy, and put a copy both into the vehicle master file at the dealer, plus into the glove box for future reference.

Assuming you have gathered the necessary information, let's go through a basic example of finding the correct part number. This works best if you follow along in your EPCnet system:

6.2. First example: Engine Oil Filter

The task is to find the correct part number for the engine oil filter. The VIN is WDB CB35 D8M A587762, which translates to a 1991 350SDL, an older S-Class Diesel. (The Vehicle Identification Charts published on the MBUSA Parts Assistance Center web site can help you decode a VIN).

1. Type in the full VIN (including the WDB manufacturer code)and press Enter. The VIN changes to the FIN (1261351A587762) in the main box, with the VIN displayed in the secondary box. Note that by entering the full VIN, most of the selections possible have already been made, the group menu opens automatically, and the Vehicle Data Card icon is active. Note that you can also select the manufacturer code manually, and then just enter the remaining characters.
2. Knowing that the oil filter is part of the engine, select the group named "Maj. Assy M

– Engine”, which is at the very top of the group list.

3. Scanning the list of Engine groups, we know that the oil filter is part of Engine Lubrication, which is Group 18, so we click on it.
4. Of the three subgroups available, the one named “Oil Filter” is the obvious choice, so we click on it, causing the illustration and parts window to open.
5. From our basic knowledge, we know that the oil filter in this model is a cylindrical object, so callout 248 seems likely. We also see callout 254, which is in a circle and has a line to callouts 245, 248, and one without a callout. A brief explanation is in order:

· Callout 254 is a kit, consisting of all the items linked by the lines. Callouts 245 (a Seal Ring or gasket) and 248 (a Filter Element) are available separately, but the third item (a ring seal of some type) is only available as a part of the Kit, callout 254. Often it is better to order the kit, since all the items necessary for a common maintenance operation will be included. In some cases, the kit will also have a small icon that looks like the end of a wrench – this denotes a Repair Kit, usually all of the gaskets and wear items.

6. If we click on callout 248, we see the part number A601 184 01 25, but it has the secondary description “Up to a reading of 1000 km”. This car isn’t new, and the odometer surely has more than 1000 kilometers (about 650 miles) on it. This item is actually the “Break-In” oil filter, and is not the right item.
7. When we click on the Kit, Callout 254, we see Part No. A601 180 01 09, along with the secondary description “From 1001 km; Range of Delivery”. Further explanation is in order:

· The first part of the secondary description is clear: Cars with more than 1001 km, which corresponds with the end of the break-in period. The second part, the term “Range of Delivery”, is seen frequently in the EPC: It means “Kit or assembly with related parts”. It usually is the main part and all or most of the other parts necessary for replacement of the main part.

8. Since Callout 254 is the clear choice, we select it by clicking on the checkbox at the left and it gets placed onto the Shopping List.

6.2.1. Basic process summary:

1. Enter VIN
2. Select Group and Subgroup
3. Select the part from the illustration
4. Select the correct part if more than one is displayed.

Not much to it, but without adequate preparation, knowledge and, most importantly, an understanding of the part, it can become quite difficult.

6.3. Second example: Front Bumper Cover

The task is to find the correct part number for a front bumper cover for vehicle WDB TJ76 H13 F033556, which is a 2003 CLK55 AMG Coupe.

1. After entering the VIN, we select Group 88 “Attachment Parts”, Subgroup 030 “Front Bumper”
2. From the illustration, the front bumper cover is callout 10, so we click on it.
3. Three possible choices appear, all similar but with different Code applications: The

first has no Code, the second has Code 494, and the third has Code 772. All three also have two footnotes, 002 and 420. Clicking on the Code Hyperlink brings up a text description stating that code: 494 is “USA Version” and Code 772 is “AMG Styling Package”. Footnote 002 states “To chassis F167229, To date 31/07/2005, Except model year 806”, and Footnote 420 states “To be painted prior to installation”.

Regarding the Footnotes, this vehicle definitely has a chassis number lower that F167229, was produced before July 31, 2005 and does not have model year code 806 (check the data card, see Section 6.5.2 for more information on model year codes and Section 6.6 for more information on Footnotes), so all three of these parts potentially apply according to the footnotes. Therefore we must also evaluate the Codes. (Note that Footnote 420 appears on a different tab from 002. Also note that the bumper must be painted).

4. Looking at the Vehicle Data Card, this car has Code 772 and Code 494, bringing up a minor dilemma: Which is the correct part? In general, you select the part that has the most applicable code(s), or if only one code is shown (as in this case), the most applicable Code. In this case, the part with Code 772 is more applicable than the part with no codes.

· In this case, one could surmise that a car with AMG Styling Package (Code 772) might have a different styling appearance for visible parts such as the bumpers, fenders and trim, as compared to the standard (Code 494) USA-version vehicles. A quick check of MBUSA Inventory shows that both the 772 bumper and the 494 bumper are in stock (meaning it is likely not incorrect), and so the correct choice is most likely A209 885 20 25. If MBUSA inventory had no record of the part, it is likely an incorrect choice.

The above example helps show why it is important to have a good understanding of automotive technology and Mercedes-Benz vehicle equipment.

6.4. Example 3: Rear Antenna

The task is to find the correct part number for the rear main radio antenna, which is located in the headliner just above the rear windshield. We’ll use the VIN WDB RF52 H16 A866863, which is a 2006 C230 Sedan.

1. Enter the VIN if necessary, it should be the same as in Example 2.
2. Radio and related equipment is in Group 82, and Subgroup 345 “Antenna” is appropriate.
3. The antenna looks like Callout 10 or 12, so click on both.
4. Callout 12 is filtered out as inapplicable, and there are three choices for Callout 10. The one with no codes can be eliminated, and the one with single codes (762/763) can also be eliminated, since the third choice has two applicable codes, both 763 (Remote Control 315 MHz) or 535 (Radio Audio 20 with CD), as well as 494 <u>and</u> 535 together. Note that there are quite a few pairs of codes listed; you have to look at all of them to find which applies. The fact that there are two codes applicable, instead of only one, makes this the correct choice in this case.

As with Example 2, again we see the importance of vehicle system knowledge, knowing that USA Version cars of this year are equipped only with the 315 MHz remote control system, never the 433 MHz version.

6.5. *Important Info for Decoding Codes*

6.5.1. Code Symbols

In Example 3 above, we saw some symbols used in the Code display: the slant bar "/" and a Plus sign "+". These symbols, along with a Minus sign "-" and Parentheses "( )" have very specific meanings, and must be interpreted correctly to understand exactly which codes do or do not apply.

· The **Slant Bar** "/" means "or". If you see a Code string of 563/645, that means "This applies to vehicles with either Code 563 OR 645"

· The **Plus Sign** "+" means "and". If you see a Code string of 563+645, that means "This applies to vehicles with both Code 563 AND 645". If the vehicle has only one of those codes, the part does not apply.

· The **Minus Sign** "-" means "without" or "not". If you see a Code string of 494 -563, that means "This applies to vehicles with code 494, and WITHOUT code 563". Regardless of all other codes, if the vehicle has 563, that part would not apply. Note that sometimes Code string "494 -563" might be written as "494 +-563". Be careful that you don't just glance and see the "+" and miss the "-".

· **Parentheses** "( )" are used to separate and group different strings to clarify meaning. For example, the Code string "494+(563/563+645/563+837+121)" means "This applies to vehicles with code 494 and one of the following: 563, or 563 AND 645, or 563 AND 837 AND 121". So, if the vehicle had Code 494, 563 and 121, but not 837, the part would not apply.

Always click on the code hyperlink to ensure display of *all* codes. Note that Codes can also appear in Footnotes, as in Example 2.

6.5.2. Some Common or Quirky Codes

· The most common code you will see is 494, "USA Version". While parts with this code are often the correct part, don't be fooled, as many parts apply to USA and non-USA cars equally and are listed next to USA-specific parts, such as in Example 2. In older models (prior to the mid-1990s), 494 means "California Version" and 491 was used to mean "USA Version". Other country codes, such as 498 ("Japan") can sometimes be used to exclude certain parts from consideration.

· The next most common codes are the three-digit "80x" codes, which mean "Model Year x". With these codes, you need to understand which ten-year span the car might be in, since Code 805 for example can mean Model Year 1985, 1995 and 2005 depending on which specific model it is being used with. To further confuse the meaning of these 80x codes, the Code description uses the "Assembly Change Year": For example, the description for Code 806 is written as "AEJ 05/1; 05/M". Even though it means "Model Year 2006" do not let the "05" fool you into thinking of Model Year 05. The assembly change year is when the assembly line changed over to that vehicle – AEJ 05/1 was around June 2005, whose vehicles would start being sold around August or September 2005, which we in the USA would refer to as Model Year 2006. These codes often appear in footnotes, where a VIN range for the part is specified with the notation "Except 80x" – even if the VIN is within the range, of the data card has Code 80x, the part doesn't apply.

· Another important code family is the four-character Trim Material codes, such as 150A or 801A. These are used like all other codes – if it's on the vehicle data card, then it applies to the vehicle – but do not confuse these with interior and exterior color codes which also have four characters (such as 151A). The trim material codes will be listed beneath the part number as a blue hyperlink, just like all other option codes, while the interior and exterior color codes are only shown within a color footnote. Both kinds of codes appear on the Data Card. Also see Section 6.6.3.

· Other codes are usually self-explanatory, or nearly so. Some codes, such as those for specific optional equipment packages, may require an understanding of the content of those packages, often available only from an old sales brochure. In most cases, however, it is only necessary to know whether the vehicle has the code on its data card or not. Some of the more tricky codes are covered in Advanced Specifying

Code strings can get quite complicated. It might be helpful to print out the code list for a particular part and check all of the possibilities against the vehicle data card. One valuable technique is to cross out or circle possibilities on a printed list of possible part numbers, eventually leaving a smaller list which, using footnotes, can be narrowed down to one, correct, part number.

6.6. Footnotes

Footnotes provide important information which will help you determine the correct part for the vehicle. The most common footnotes provide guidance when a part number application has changed during the production run. Other footnotes will lead you to the correct color code for a part, or other information necessary to specify the part.

6.6.1. VIN breakpoint footnotes: From Chassis No. / Date; Up To Chassis No. / Date

In the course of vehicle design and manufacture, sometimes part designs change over time. The reasons for this are varied, and include cost efficiencies, part sharing between models, and styling updates. The VIN / Date breakpoint footnotes guide you to the correct part for that specific vehicle. While usually straightforward in application – if the footnote states “Up to A123456” and your vehicle’s VIN is A345678, then the part clearly does not apply. However, be cautious when the VIN and breakpoint are near each other, for example if your vehicle’s VIN was A123555 – that is, when the vehicle is close in either VIN or date to the breakpoint - it is a good idea to verify the correct part as installed on the vehicle. The reason for this is that vehicle production does not always occur in strict numerical order, with some cars having lower VINs leaving the production line after others with higher VINs

Date footnotes work in much the same way, with the important distinctions that they are often more accurate than a VIN breakpoint, and that the dates are written in World format – DD.MM.YY. This means that the date 05.12.04 (sometimes written as 05/12/2004) is December 5, 2004. In some cases, only the Date is given. If a footnote breakpoint for your vehicle’s plant code is not given, the breakpoint can be approximated from that for the other plant codes. Although the production plants will phase in a new part at slightly different times, they all happen at about the same time.

Sometimes the footnote will exclude or include vehicles based on a Data Card Code, such as the “80x” codes (see Section 6.5.2). Even if the VIN is within the footnote range, if the footnote also states “Except 80x”, and the Data card has 80x, then the part is not valid for that vehicle.

6.6.2. To be adapted during assembly

As with the example above, parts sometimes change over time. In some cases, the technician must make the part supplied fit the vehicle by modifying it slightly with common tools. A trim piece may need to be shaved to fit exactly, or an extra hole drilled for mounting, or a rubber seal or hose may need to be cut to length. In such cases, the part supplied will not fit the vehicle as it is; it is important to advise the technician of this.

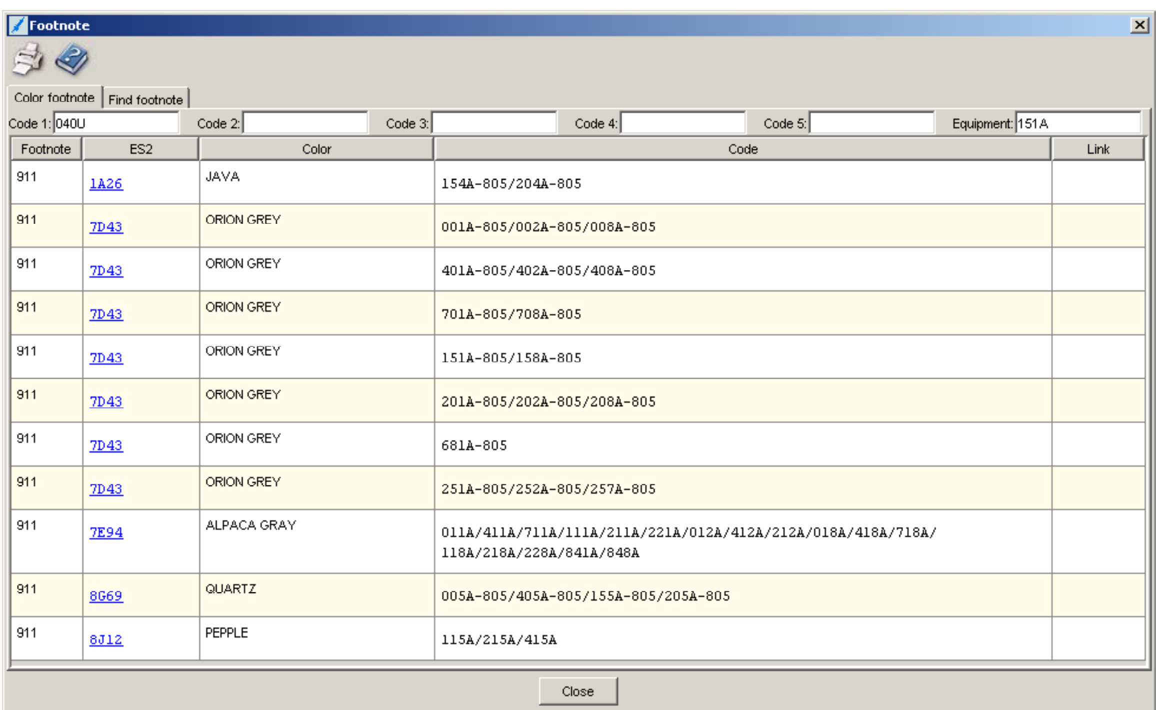
6.6.3. Color Footnote

As explained in Sections 4.4 and 6.5.2, a four character ES2 code is often used to designate the specific color of a part. To identify the correct ES2 color code for your vehicle, simply match up the vehicle's color designation code with the part's color code on the list in the footnote. In the example below, which is for an interior trim piece, the vehicle's interior trim color code (shown at the upper right) is 151A. Looking in the Code column for 151A, about half way down we find the correct ES2 for this code to be 7D43, unless this car was a Model Year 2005 (minus code 805), which it isn't. Note that the EPC will in many cases filter out the ES2 codes that do not apply – the example below has all the filters switched off.

Codes that look like a color code, but are not, are used to show the material from which the part is made. This is particularly evident when specifying items such as seat coverings and inner door panels, which may constructed from materials such as Leather, Nappa Leather, MB Leather-Twin (a synthetic leather, also called MB-Tex or Tex-Leather), and Fabric. These codes are shown on the Vehicle Data Card, but are not the Trim Color Code. For the example above, the code 150A also appears on the data card, denoting MB Leather-Twin. The EPC uses this code to filter out part choices which do not apply. Note that in many cases, a part may apply to two or more types of material, such as 150A and 700A.

Also, the secondary text of a part number listing contains important information for specifying the part. Codes are important, but so is the part description.

Finally, be aware that any vehicle with a Designo color uses the data card option code colors (like Y30), not the standard color code (like 151A). This is a very common source of confusion and incorrect parts. Often in these cases there will be a link in the right-most column (the image below shows none) – follow that link for important information.



Footnote	ES2	Color	Code	Link
911	1A26	JAVA	154A-805/204A-805	
911	7D43	ORION GREY	001A-805/002A-805/008A-805	
911	7D43	ORION GREY	401A-805/402A-805/408A-805	
911	7D43	ORION GREY	701A-805/708A-805	
911	7D43	ORION GREY	151A-805/158A-805	
911	7D43	ORION GREY	201A-805/202A-805/208A-805	
911	7D43	ORION GREY	681A-805	
911	7D43	ORION GREY	251A-805/252A-805/257A-805	
911	7E94	ALPACA GRAY	011A/411A/711A/111A/211A/221A/012A/412A/212A/018A/418A/718A/ 118A/218A/228A/841A/848A	
911	8G69	QUARTZ	005A-805/405A-805/155A-805/205A-805	
911	8J12	PEPPLE	115A/215A/415A	

6.6.4. First Exhaust Stock of Old Parts up to Ident No.

This footnote is advising you to order the earlier version part, up to a certain chassis or VIN break point, instead of ordering the later number in a supersession. If the part is available, it is often a better fit or form than the replacement part and should be used.

6.6.5. The Old Part Must No Longer Be Installed

Unlike the advice given above in section 6.6.4, when you see this footnote, you should avoid installing the previous version part. This is a good time to check your inventory; if the old part is in stock, it may be advisable to scrap, or return these parts with a note to the effect "EPC states these parts are not to be installed". There are some infrequent exceptions which are made clear through NetStar News Channel Announcements.

6.6.6. Part is Only or Also Supplied as Reconditioned Part

This footnote is advising you to check for a remanufactured version. The EPC generally does not list parts with an ES2 of 80 (designating a remanufactured or rebuilt part). In general, the remanufactured part is less expensive, after considering the return of the core value, than the new part, and may be the only part available.

6.6.7. When Ordering, ... (Quote VIN or ES1 26 or...)

In some cases, a part must be specially prepared before delivery to the dealer, to ensure it will function properly in the specific vehicle. Some common examples of this requirement are keys, locks, and some electronic control modules. There may be other parts requiring a VIN or specific ES1 code (sometimes called an *Index*) for ordering, even though no footnote is shown in the EPC – this is managed in the parts ordering system (i.e., Paragon). If the footnote directs you to order with a specific ES1 code, failing to do so will generally result in receiving an unprepared part which will not function properly in the vehicle.

6.6.8. When selling brake linings...

This footnote, shown at all brake lining, pad and shoe parts, is reminding you that brake maintenance presents some risk to the user from fine dust particles. Workshop technicians are trained to protect against these risks with proper protective equipment and procedures, but such training cannot be assumed in the case where a retail customer purchases parts over the counter. Be sure to advise the customer of risks associated with these parts. If desired, the multi-language brochure referenced in the footnote may be given to the customer, which advises of these risks. A photocopy of the appropriate language may be provided, or the brochure itself, but in all cases this must be provided to the customer free of charge.

6.6.9. Order by the meter

Some materials are provided as so-called 'meterware', in that it is sold by the meter (about 39 inches). Thus ordering a quantity of 10 will result in receiving 10 meters of that item, about 32 feet. In many cases, the length of material required for a specific job or application will be stated in the EPC near the part number. If a part is shown as being 1330 mm (1.33 meters) long, then it be necessary to order at least 2 meters for the job. Common items sold by the meter are smaller hoses, tapes, felt strips and the like.

6.6.10. Note Cond. Return Via Syrek Settlement in Germany

This footnote and similar ones apply only to dealerships within Germany or other specified countries. It is a way of alerting these dealers to return the replaced part via the Syrek system for warranty inspection, review, remanufacturing, recycling or other processing. Follow the appropriate return instructions for your country.

6.6.11. Other important messages (exchange in pairs, always also install xxx, etc.)

Not every footnote can be explained in this document. Therefore, it is important to read and understand each footnote – they contain important information. If you don't understand what the footnote is trying to say, ask for help at your dealership, or from the MBUSA support team.

6.6.12. Contact Sizes & Spring Values (covered in Advanced)

In some cases, a footnote must be interpreted to select the correct part from several choices. Two common examples are the correct size of wire splice, which depends upon the total cross-sectional area of the wires to be joined, and the selection of suspension springs and shims, which depend upon vehicle equipment. These footnotes are covered in Advanced Specifying, and (as with all footnotes) are critically important for identifying the correct part for the vehicle.

6.7. Supersessions and Substitutions

The EPC sometimes shows a supersession or substitution. This is done when the original part is no longer available, or a newer type part is superior in some way. It can also simply mean that the later version part is what can be supplied, and may be the only possibility for a repair. If the prior version part is available, it is often the best choice to repair the vehicle.

6.7.1. Supersession with additional parts

In some cases, the superseding part does not exactly replace the superseded part, instead it must be modified, or other parts must also be replaced, in order for the part to have the correct fit, form or function. Frequently, these additional parts are necessary only for certain vehicles.

In the parts window, the original part will be shown as 'Replaced By" some other number. If there is a footnote or code associated with the original listing, plus additional parts are shown, it means that vehicles meeting the criteria of the footnote or code must also use the additional parts for the repair.

In the image below, vehicles "Up To Engine No. ..." not only need the A606 200 00 22, they also need those additional parts. In such cases, have the technician inspect the vehicle: It is possible that this same repair was performed earlier, and the additional parts are already installed.

✓	Item no.	Part number	Designation/description	Quantity	Version
☐	248	A 603 200 04 22	**FAN DRIVE** ☐ **Replaced by:** A 606 200 00 22 +001 A 606 200 00 23 +003 N 914020 006017 [021] Up to engine:	-	
☐	248	A 606 200 00 22	**FAN DRIVE** [022] As of engine	001	

6.7.2. General Supersessions

Again, it is important to understand that a supersession may not necessarily be a good thing, nor is it necessarily a bad thing. Parts are superseded for many reasons, and quality or performance is rarely the reason. The most common reason for a supersession is *Assortment Simplification*: reducing the number of different parts that must be kept in stock. For ease of production, there may have been five versions of a part, while as a spare part, one version will fill the need, thus all versions are replaced by a single version.

Generally, such supersessions are managed by the parts ordering system (such as Paragon), so it is often in your best interest to order the part number that was removed from the vehicle, allowing the parts ordering system to deliver the part(s) determined to fulfill that need. This will not work in every case, but it is a good policy in general. The primary source of supersessions is the parts ordering system, not the EPC.

6.8. Illustrations

When working with the EPC, illustrations are a critical part of the information used to specify parts. As such, you will often encounter symbols in the illustrations. It is important to understand what these symbols are telling you.

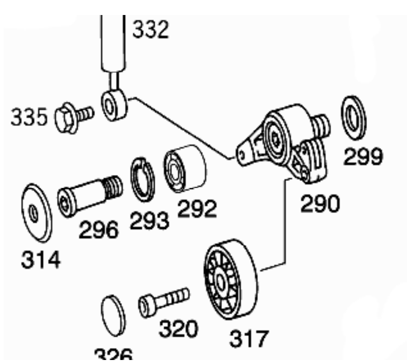
6.8.1. General Information

Illustrations do not always show the part exactly as it appears in real life. The drawings, while often fairly accurate, are intended to be generalized and not necessarily an exact rendering. The illustration may show more than one version of a part, without anything to indicate the differences. For example, an ashtray is available in two versions, early and late production. The ashtray may be shown as a single drawn item, or may be shown as two separate items, possibly on two separate image pages.

In practical use, this means that you need to look at the images, the part description and secondary text, the footnotes, the codes, and all other information provided before coming to a conclusion. If you click on a part image, and see the message "Item No. Not Valid for this Model Designation", it may mean that you need to look just a little bit further.

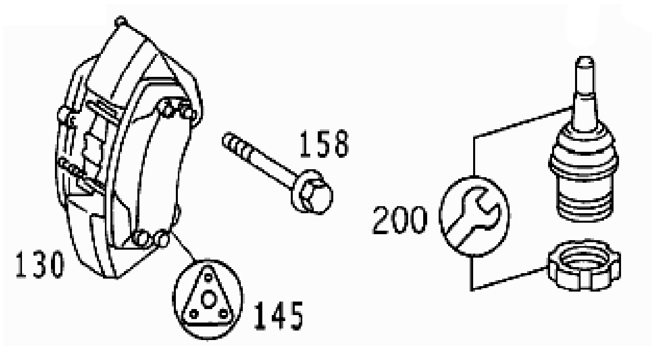
6.8.2. Lines and Arrows

One very common symbol is a line. Lines are used to show that something fits into something else, or where it fits into the assembly. In every case, some common sense will ensure the meaning of the line is clear. Arrows are also used in the way, to enhance understanding of a drawing.



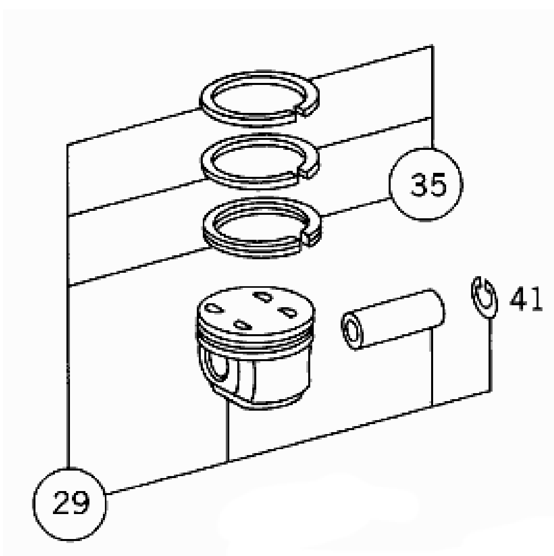
6.8.3. Repair Kit & Gasket Kit

The two symbols shown the in the image below are very common. The triangular symbol with the holes denotes a Gasket Kit, which includes all relevant gaskets for an assembly. The Wrench symbol denotes a repair kit, which contains the parts to rebuild or repair an assembly. In general, you would order the gasket kit if the part is leaking, and the repair kit if there was some other functional fault. If the item is damaged or broken, the entire assembly may be required.

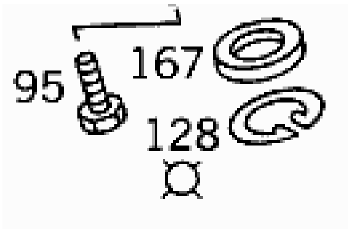


6.8.4. Kit of Parts

The circle symbol, along with the associated callout and lines shows the contents of a Kit of Parts. When you order the part number designated by the callout number, you will receive all of the items designated by the lines. For example, 29 includes the rings, piston, wrist pin and wrist pin clip, while 35 is only the rings. The wrist pin clip is available separately as Item 41.

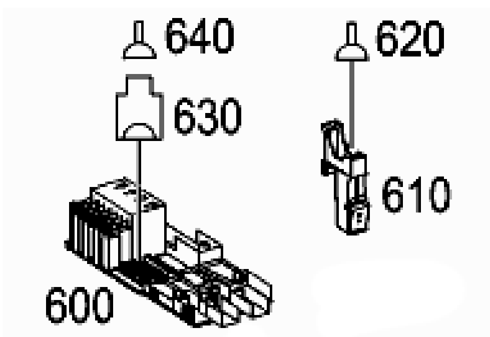


When a Kit of Parts has components shown over a large area of the illustration, or on more than one illustration, contents of the kit are shown with the circle symbol shown just below callout 128.



6.8.5. Connectors

In older catalogs, electrical connectors are shown much as they appear. In newer catalogs, this has been replaced by a pair of symbols, one for the connector body or housing, and one for the electrical contact. In the image below, 630 is the housing, and 640 and 620 are electrical contacts.



6.8.6. Software

A symbol of a Compact Disk (CD-ROM) is used to denote software, used to program or reprogram a part (usually a control unit). Although a part number may be provided, in general software is not offered or delivered through the regular spare parts system. Instead, the Service Department should use their Star Diagnosis System (SDS) or Diagnosis Assistant System (DAS) equipment to obtain and load the software.

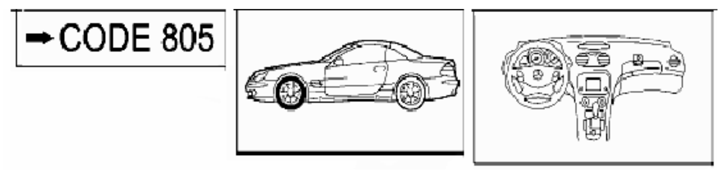
6.8.7. Reference to Another Group

In some cases, to ease your task of finding the parts you need, a reference to another group and subgroup. If you see this symbol, and the part you need either is not shown, or you need more information, you should have a look in the group and subgroup shown.

6.8.8. Applicability Note

In some cases, information as to the applicability of the illustration is shown in the upper right corner. Generally these are symbols, so as to remain language neutral. You might see an engine type (M113 for example), which means if you're looking for parts for an M642 engine, keep looking until you find the correct illustration. Front and Rear brake parts might be shown this way, with a picture of a car and the front (or rear) wheels drawn in bold. Model year applicability (such as "805 >" which means 2005 [*or 95 or 85, etc*] and later), model applicability (such as "211"), an optional suspension system (Hydropneumatic, Airmatic, etc), and many other hints are given here.

The three images below mean "Up To Code 805", "Front Axle", and "Left Hand Drive" respectively.



It is important to take note of these symbols at all times, otherwise you risk looking for parts in the wrong place.

6.9. Conclusion

With an understanding of this information, you should be ready to specify most parts. However, learning and mastering are two different things, and mastery takes practice and good sense. Not remembering what you learned defeats mastery.

Every Mercedes-Benz Parts Specifier was in this situation at one time, and the majority learned, one way or another, how to specify parts. When you are specifying parts, and things are not going well, ask for help. Your colleagues at the parts counter can often add some expertise to the situation. The technician is your ally in locating the right part, since they likely have a good idea of what it is, where it fits, and how it works. Your parts manager or Foreman should be willing to assist when necessary. Mercedes-Benz USA offers the services of Dealer Parts Services for dealership personnel.

Eventually, every part number is found. Persistence, common sense, and continuous learning will serve you well.

Appendix A. Advanced Parts Specification

This section of the guide is somewhat less structured than previous sections, as it deals with many different specific issues you may encounter while specifying. This material cannot be learned in a few days; in fact, it may be months or years before this material is mastered. This information represents the accumulated knowledge and experience of dozens of the best parts specifiers worldwide. While it is meant to be a detailed guide, it is by its nature incomplete, with the gaps filled only through experience. MBUSA welcomes your comments for additions and enhancements to future editions of this section of the guide.

A.1. Specifying Suspension Springs

One of the more confusing tasks in specifying is finding the correct suspension springs. While the process is not overly complex, there are many opportunities for error. In general, the process goes like this:

1. Print out vehicle data card (This is optional, but eases the process significantly)
2. Locate spring illustration and select an appropriate part listing
3. Open the footnote, and print it. (Again, optional, but strongly recommended)
4. On the footnote printout, identify the option codes for the vehicle
5. Add together the base points for the model, plus the points for the options.
6. Identify the correct column and locate the entry for the point value calculated
7. Read the part number necessary, along with the spacers if necessary

By now, the first three steps should be well understood, so we will focus on the last four steps. We'll use the VIN WDB RF52 H16 A866863, which is a 2006 C230 Sedan, and specify the rear springs. In most cases, you'll want to replace the springs in pairs, but that is a decision for the Service department. Navigating to Group 32, subgroup 120, there are two obvious choices, since this vehicle has 486+494+949. It doesn't matter in practice which one you choose, as long as the footnote number (061 in this case) is the same. Clicking on the footnote link, we see the footnote (shown on the next page):

The option codes (and associated points) on the data card that match those in the footnote are:

275 (1)

414 (3)

427+203.052 (1)

540 (1)

690 (4)

We also find that the base points for 203.052 are 166, so the total number of points equals 176. It is important to get this correct, so double-check your work, or even better, have someone else check your work.

Looking at the parts tables lower in the footnote, we find only one that has all the applicable codes: "Sports Sus. USA Sports Pack. 'Evolution' Code 486+494+949". Note that other possible entries, such as "494 USA" and "(494/494+486)+M005" do not apply, the first because there's another entry with more applicable codes, and the second one because this vehicle does not have M005.

Text footnote | Find footnote

```
061  TO DETERMINE THE REAR SPRINGS, THE POINTS IN THE TABLE BELOW   FOR EACH RESPECTIVE MODEL AND THE SPECIAL VERSIONS
061  INSTALLED ARE TO BE ADDED UP
061  ============================================================== ============================================================
061  MODEL     POINTS TOTAL                CODE            SPECIAL  EQUIPMENT         POINTS TOTAL
061  203.004        165                     248            R. SEAT  W. INTEG. CH. SEAT     1
061  203.006        166                     275            DRIVER.  SEAT, ELECTRIC          1
061  203.007        165                     282            SKI BAG  1
061  203.008        166                     287            THROUGH  -LOADING FEATURE        1
061  203.016        169                     352            COMMAND  1
061  203.018        174                     413            FULL GL  ASS ROOF                4
061  203.020        169                     414            SLIDING  ROOF IN GLASS           3
061  203.035        163                     423            TRANS.,  5-SPEED, AUTOMATIC..    1
061  203.040        166       427+203.052/054/056          TRANS.,  AUTOMATIC, 7-SPEED..    1
061  203.042        166       427+203.020                  TRANS.,  AUTOMATIC, 7-SPEED..    2
061  203.043        166                     474            PARTICU  LATE FILTER             1
061  203.045        166                     480            LEVEL.   CONTROL....             2
061  203.046        166                     540            REAR WI  NDOW ROLLER BLIND, E..  1
061  203.052        166                     550            TRAILER  HITCH......             8
061  203.054        160                     550+669        SPARE..  WHEEL                   7
061  203.056        167                     669            SPARE..  WHEEL, FOLDABLE         6
061  203.061        166                     679            STRUCTU  RAL WHEEL AS SP. WHL.   4
061  203.064        167                     690            C. SP.   TIRE                    4
061  203.065        174                     810            SOUND S  YSTEM                   1
061  203.076        173                     970            POLICE.  6
061  203.081        166                     970+962        POLICE.  + ROOF EXT. HELLA.      12
061  203.084        167                     974            FIRE...  DEPT.                   5
061  203.087        167                     974+962        FIRE DE  PT. + ROOF EXT. HELLA   11
061  203.092        166                     975            E. PHYS  5
061  975+962        E. PHYS                                         +ROOF EXPAN., HELLA     11
061  ==========================================================================================================================
061            REAR SPRINGS TABLE/RUBBER LININGS
061  ==========================================================================================================================
061            SUM OF POINTS                REAR SPRINGS                  BOOT
061  STANDARD SUSPENSION AND 494 USA
061   AND 498 JAPAN         -145                        202 324 17 04210 325 01 04
061       146-152                      202 324 17 04            210 325 02 84
061       153-159                      202 324 17 04            210 325 03 84
061       160-166                      202 324 17 04            210 325 04 84
061                      LARGE GRND CLEARANCE                 CODE 482
061       167-171            -155      202 324 18 04            210 325 01 84
061       172-178         156-162      202 324 18 04            210 325 02 84
061       179-185         163-169      202 324 18 04            210 325 03 84
061       186-191         170-176      202 324 18 04            210 325 04 84
061  *
061       192-197         177-181      202 324 19 04            210 325 01 84
061       198-204         182-188      202 324 19 04            210 325 02 84
061       205-211         189-195      202 324 19 04            210 325 03 84
061       212-250         196-250      202 324 19 04            210 325 04 84
061  ==========================================================================================================================
061  TABLE REAR SPRINGS - RUBBER BOOT "SPORTS SUSPEN." CODE 486..  OR "SPORTS SUS." USA SPORTS PACK. "EVOLUTION" CODE 486+494+949
061  ==========================================================================================================================
061            SUM OF POINTS                REAR SPRINGS                  BOOT
061         -150                       203 324 05 04            210 325 01 04
061       151-155                      203 324 05 04            210 325 02 84
061       156-165                      203 324 05 04            210 325 03 84
061  *
061       166-173                      210 324 34 04            210 325 01 84
061       174-181                      210 324 34 04            210 325 02 84
061       182-188                      210 324 34 04            210 325 03 84
061       189-250                      210 324 34 04            210 325 04 84
061  M005 (NORMAL SUSPEN. & USA SPORTS CHASSIS) / (494/494+486)+M00.5
061                                                             SHIM...
061         -152                                                210 325 01 84
061       153-159                                               210 325 02 84
061       160-166                                               210 325 03 84
061       167-171                                               210 325 04 84
061       172-178                                               210 325 01 84
061       179-185                                               210 325 02 84
061       186-191                                               210 325 03 84
061       192-197                                               210 325 04 84
061       198-204                                               210 325 01 84
061       205-211                                               210 325 02 84
061       212-218                                               210 325 03 84
061       219-250                                               210 325 04 84
061  AMG M112+M001 203.065
```

061	219-250	210 325 04 84
061	AMG M112+M001 203.065	
061	INSERT.	
061	-181	210 325 01 84
061	182-189	210 325 02 84
061	190-250	210 325 03 84
061	AMG USA M112+M001+494 203.065	
061	INSERT.	
061	-180	210 325 02 84
061	181-250	210 325 03 84
061	M612+M30 203.018	
061	INSERT.	
061	-183	210 325 01 84
061	184-190	210 325 02 84
061	191-198	210 325 03 84
061	199-250	210 325 04 84
061	M113+M55 203.076	
061	INSERT.	
061	-181	210 325 01 84
061	182-189	210 325 02 84
061	190-250	210 325 03 84
061	M113+M55+494 (USA) 203.076	
061	INSERT.	
061	-180	210 325 02 84
061	181-188	210 325 03 84
061	189-250	210 325 04 84
061	M113+M55 & M113+M55+494 203.076	
061	REAR SPRING	
061	210 324 34 04	
061	M112+M001 & M112+M001+494 203.065	
061	REAR SPRING	
061	210 324 34 04	
061	M612+M30 203.018	
061	REAR SPRING	
061	210 324 34 04	
061	--	

Lastly, finding the entry in the correct part of the table for 176 points, we see that A210 324 34 04 are the correct rear springs, and A210 325 02 85 are the correct "linings" or shims.

Note that, in some cases (particularly in vehicles 12/95 and earlier) there are rubber or metal shims that must also be ordered, available in 5mm height increments. These will be shown in the EPC, and must be ordered so as to make the two springs plus shims to be as similar as possible in height. Note that some springs may be marked on the last coil with a colored paint dot. As explained in WIS, the paint color indicates the height tolerance for that particular spring. Note that paint dots appearing on coils <u>other than</u> the first one are used only in the manufacturing process and have no meaning for spare parts.

A.2. Model 209 CLK Soft Top Assembly

While uncommon, the selection of the correct soft top assembly color combination in a Model 209 is a challenge. Much like the selection of springs above, this is an exercise in patience and code resolution.

A typical footnote for this part is shown below and on the next page, as it would appear on the screen. Printing it helps, but in some cases the columns no longer line up. The trick here is to cross off all of the entries with soft top exterior color codes (740, 744 or 749) that do not apply, then to look at which interior codes (such as 817A) that do apply. From the short list obtained, then review each one to select the one where the remaining codes match the vehicle equipment – codes such as 806, or -Y83, and the VIN range.

Again, it becomes a process of elimination: Pick a code type to eliminate many choices quickly, then repeat that process, splitting the list into smaller and smaller pieces until only one part – the correct one – remains.

Ident. end no. | Text footnote | Find footnote

```
900   ==========================================================================================================
900   I    0101 I                                        I 740+822A-Y83/740
900   I         I                                        I +827A-Y83/740+828A
900   I         I                                        I -Y83/740+831A-Y83
900   I         I                                        I /740+832A-Y83/740
900   I         I                                        I +838A-Y83/740+858A
900   I         I                                        I -Y83/740+848A-Y83
900   UP TO IDENT:          T 052301    I    0101 I                                  I 251A+740-806-Y83
900   UP TO IDENT:          F 168787    I    0101 I                                  I /252A+740-806-Y83
900   I         I                                        I /257A+740-806-Y83
900   I         I                                        I /431A+740-Y83/731A
900   I         I                                        I +740-Y83
900   I    0102 I                                        I 749+822A-Y83/749
900   I         I                                        I +827A-Y83/749+828A
900   I         I                                        I -Y83/749+831A-Y83
900   I         I                                        I /749+832A-Y83/749
900   I         I                                        I +838A-Y83/749+858A
900   I         I                                        I -Y83/749+848A-Y83
900   UP TO IDENT:          T 052301    I    0102 I                                  I 251A+749-806-Y83
900   UP TO IDENT:          F 168787    I    0102 I                                  I /252A+749-806-Y83
900   I         I                                        I /257A+749-806-Y83
900   I         I                                        I /431A+749-Y83/731A
900   I         I                                        I +749-Y83
900   I    0104 I                                        I 744+822A-Y83/744
900   I         I                                        I +827A-Y83/744+828A
900   I         I                                        I -Y83/744+831A-Y83
900   I         I                                        I /744+832A-Y83/744
900   I         I                                        I +838A-Y83/744+858A
900   I         I                                        I -Y83/744+848A-Y83
900   UP TO IDENT:          T 052301    I    0104 I                                  I 251A+744-806-Y83
900   UP TO IDENT:          F 168787    I    0104 I                                  I /252A+744-806-Y83
900   I         I                                        I /257A+744-806-Y83
900   I         I                                        I /431A+744-Y83/731A
900   I         I                                        I +744-Y83
900   I    0201 I                                        I 740+835A-Y83
900   I    0202 I                                        I 749+835A-Y83
900   I    0204 I                                        I 744+835A-Y83
900   UP TO IDENT:          T 052301    I    0401 I                                  I 740+Y83-806
900   UP TO IDENT:          F 168787    I    0401 I                                  I
900   UP TO IDENT:          T 052301    I    0402 I                                  I 749+Y83-806
900   UP TO IDENT:          F 168787    I    0402 I                                  I
900   UP TO IDENT:          T 052301    I    0404 I                                  I 744+Y83-806
900   UP TO IDENT:          F 168787    I    0404 I                                  I
900   FROM IDENT:           T 048983    I    1101 I                                  I 251A+740+806-Y83
900   FROM IDENT:           F 132632    I    1101 I                                  I /252A+740+806-Y83
```

900	FROM IDENT:	F 132632	I	1101 I		I /252A+740+806-Y83
900	UP TO IDENT:	T 052301	I	1101 I		I /257A+740+806-Y83
900	UP TO IDENT:	F 155084	I	1101 I		I /411A+740-Y83/711A
900	I I				I +740-Y83	
900	UP TO IDENT:	T 052301	I	1101 I		I 740+811A-Y83/740
900	UP TO IDENT:	F 155084	I	1101 I		I +818A-Y83/740+814A
900	I I				I -Y83/740+817A-Y83	
900	I I				I /740+824A-Y83	
900	FROM IDENT:	T 048983	I	1102 I		I 251A+749+806-Y83
900	UP TO IDENT:	T 052301	I	1102 I		I /252A+749+806-Y83
900	I I				I /257A+749+806-Y83	
900	I I				I /411A+749-Y83/711A	
900	I I				I +749-Y83	
900	UP TO IDENT:	T 052301	I	1102 I		I 749+811A-Y83/749
900	I I				I +818A-Y83/749+814A	
900	I I				I -Y83/749+817A-Y83	
900	I I				I /749+824A-Y83	
900	FROM IDENT:	T 048983	I	1104 I		I 251A+744+806-Y83
900	FROM IDENT:	F 132632	I	1104 I		I /252A+744+806-Y83
900	UP TO IDENT:	T 052301	I	1104 I		I /257A+744+806-Y83
900	UP TO IDENT:	F 155084	I	1104 I		I /411A+744-Y83/711A
900	I I				I +744-Y83	
900	UP TO IDENT:	T 052301	I	1104 I		I 744+811A-Y83/744
900	UP TO IDENT:	F 155084	I	1104 I		I +818A-Y83/744+814A
900	I I				I -Y83/744+817A-Y83	
900	I I				I /744+824A-Y83	
900	FROM IDENT:	T 052302	I	1111 I		I 251A+740/252A+740
900	FROM IDENT:	F 155085	I	1111 I		I /257A+740/711A+740
900	FROM IDENT:	T 052302	I	1111 I		I 740+811A/740+815A
900	FROM IDENT:	F 155085	I	1111 I		I /740+818A/740+814A
900	I I				I /740+817A/740+824A	
900	FROM IDENT:	T 052302	I	1112 I		I 251A+749/252A+749
900	I I				I /257A+749/711A+749	
900	FROM IDENT:	T 052302	I	1112 I		I 749+811A/749+815A
900	I I				I /749+818A/749+814A	
900	I I				I /749+817A/749+824A	
900	FROM IDENT:	T 052302	I	1114 I		I 251A+744/252A+744
900	FROM IDENT:	F 155085	I	1114 I		I /257A+744/711A+744
900	FROM IDENT:	T 052302	I	1114 I		I 744+811A/744+815A
900	FROM IDENT:	F 155085	I	1114 I		I /744+818A/744+814A
900	I I				I /744+817A/744+824A	
900	UP TO IDENT:	T 052301	I	1201 I		I 740+815A-Y83
900	UP TO IDENT:	F 155084	I	1201 I		I
900	UP TO IDENT:	T 052301	I	1202 I		I 749+815A-Y83
900	UP TO IDENT:	T 052301	I	1204 I		I 744+815A-Y83
900	UP TO IDENT:	F 155084	I	1204 I		I
900	I 1221 I				I 740+805A	
900	I 1222 I				I 749+805A	
900	I 1224 I				I 744+805A	
900	FROM IDENT:	T 048983	I	1401 I		I 740+806+Y83
900	FROM IDENT:	T 048983	I	1402 I		I 749+806+Y83
900	FROM IDENT:	T 048983	I	1404 I		I 744+806+Y83
900	FROM IDENT:	T 052302	I	1441 I		I 411A+740/740+801A
900	FROM IDENT:	F 155085	I	1441 I		I /740+808A
900	FROM IDENT:	T 052302	I	1442 I		I 411A+749/749+801A
900	I I				I /749+808A	
900	FROM IDENT:	T 052302	I	1444 I		I 411A+744/744+801A
900	FROM IDENT:	F 155085	I	1444 I		I /744+808A
900	==					

A.3. Keys & Locks

Ordering Keys and Locks can be confusing, however there are some external aids which can be used to ensure the correct part is ordered. Although there are documented cases where the key code database is incorrect for a specific vehicle, this is rarely the problem when keys don't work. Far more often either the wrong key has been ordered, an incorrect "learning" or "synchronizing" procedure has been used, or the locking system has a fault that has not been properly diagnosed.

There are Dealer Technical Bulletins and Service Informations published which address issues relevant to specific models and years. These publications should be reviewed to gain familiarity. Additionally, MBUSA publishes a "Key Chart" which can assist with the specification of keys.

A.3.1. Mechanical Keys and Blades

Mechanical keys, and key blades for electronic keys, are cut to the key code on file for the specified VIN. These are simply pieces of metal, cut to shape. Although it is important to start with the correct key blank, and have the correct lock code, ordering and supply is straightforward. Certain key styles, particularly the old alarm system keys with a red insert, and the so-called 'flat' or 'wallet' key, are no longer available. In the first case, the red insert has no function, and so the replacement key looks and functions the same, just the red insert is missing. For the flat key, simply remove the plastic head of a standard key, revealing the flat metal part. In the rare case a lock has been changed with an undocumented key code (e.g. from a recycled vehicle), it may be possible in some cases to decode the mechanical key.

A.3.2. Electronic keys: Non DAS

Starting in Model Year 1990, with the R129 SL Class, Mercedes-Benz started using electronic remote keys. Until the Drive Authorization System (DAS, sometimes called FBS) was introduced around 1996, there was no limit to the number of keys a customer could order or have – all keys were identical.

In cars with electronic keys, there are two lock codes: Mechanical and Electronic. These two codes are independent of each other. The mechanical code may be shown on the vehicle data card, but the electronic code is secret. It is not possible to determine the electronic code by decoding a key or control module.

Once an electronic key has been programmed, it is impossible to re-program it for another vehicle. Electronic keys can only be ordered pre-programmed for a specific VIN

A.3.3. Electronic keys: DAS

With the introduction of various versions of Drive Authorization Systems, keys became individualized and carried specific information, such as driver seat position settings. For most vehicles, there can be a maximum of eight (8) active keys at any moment. Each of these 8 keys can be replaced up to two times, for a lifetime total for the vehicle of 24 keys. If for some reason a customer needs a 25th key, the entire locking and DAS system (which includes the engine control module and/or the shifter/steering column module) will have to be replaced.

There is one notable exception to the 24 key limit: Model Year 1998 and 1999 W163 M-Class vehicles: For these vehicles <u>only</u>, the lifetime maximum is eight keys total.

When specifying keys for vehicles equipped with a Drive Authorization System, it is necessary to differentiate between a Replacement Key and an Additional key. A Replacement is used to replace an existing key, and the specific key number (read via Star Diagnosis) must be specified. An Additional Key is just that, a new key in addition to those already existing, and an unused key number must be specified. Specify Replacement or Additional, plus the key number, in the turnaround field when placing the order.

A.3.4. Control Modules

Numerous versions of the locking and DAS systems bring with them many versions of control module. In newer DAS3 models – those with the SmartKey – the main control module is the Electronic Ignition Switch (EIS). When ordering an EIS, a separate **programming key** must also be ordered at the same time. In other models, there is often a remote unlocking module, and a separate DAS module. Make sure you know which one you need to order, as it's easy to order the wrong one. In the M Class, 1998 and 1999 models use the All-Activity Module (AAM) as the controller, later years use a different module named FBM as the controller.

A detailed explanation of the many systems, their components, and functions, is beyond the scope of this document. However, this information is available in the service literature, such as WIS, Wiring Diagrams, and service bulletins. An experienced shop foreman or technician may also be of assistance.

A.3.5. Complete new lock sets

MBUSA can supply complete new lock sets for most models. Note that this does not include the electronic coded portion of a locking system: In general, the electronic code number is assigned for the life of the vehicle. It is never necessary to replace an electronic code, for example in case of theft of a key, since any electronic key can be disabled in the workshop.

Complete lock sets are blocked and can only be ordered through Dealer Parts Services. It is important to confirm the installation of a lock set that alters the vehicle key code, so the records can be updated. Also note that the service department must complete a form stating that the complete lock set is to be installed only at a dealership; lock sets cannot be given to another service facility or directly to a customer for installation.

A.3.7. Workshop locks

Some lock cylinders can be ordered with a so-called workshop key. This is a lock with a randomly-assigned key code. These parts are intended to be used as a 'loaner' for a faulty component in a customer car, to allow the vehicle to be driven while waiting for the special-ordered part to arrive. For example, if the ignition switch tumbler is intermittently jamming, installing the tumbler from the Workshop lock set will allow the customer to be on his way, using a different (workshop) key for the ignition, until the new tumbler for that VIN arrives.

A.4. Wire Splices

A table for wire splices is often presented as a footnote. The task here is to add up the combined cross sectional areas of all of the wires to be spliced together, and then order the splice that will allow those wires to be joined properly. The service department needs to identify which wires are being joined by looking at both the car and the wiring diagram, after which the combined cross-sections of all the wires is easily calculated.

The reason for wire splices to exist in the first place is that certain 'repair harnesses' are available. Instead of replacing the entire harness for a damaged connector, the repair harness is spliced in to the harness, providing for a reliable, durable and safe repair.

For these wire splices, it is a good idea to keep an assortment in stock.

A.5. More Codes

264 – License Plate Attachment Americas. Most newer USA-version cars have this option, but it may not be listed on the Vehicle Data card.

772 – AMG Version. It is important to note whether this code is on the Vehicle data Card or not, since many choices are determined or ruled out by this code. Note that just because there may be a part stating "772" there is not necessarily a part stating "-772" also – in some cases the standard part applies.

A.6. Info Records

For a very few select vehicles, after entering the VIN you will see a window with a lengthy message, sometimes in German. Try this with Chassis Number WDD 219.376-1A-060747. Whenever the production plant builds a vehicle that is special in some way, it is documented with a so-called *production information record*. This document records the information necessary to specify certain parts for this vehicle specifically, and absolutely must be followed in order to obtain the correct parts.

In the example VIN given, this is one of 55 "IWC Ingeneur" AMG versions that were built for MBUSA in 2005. About a hundred of the trim parts, both interior and exterior, were specially finished or painted. Ordering the standard part will result in a mismatched part. For assistance in translating this somewhat technical German, if your dealership does not have someone fluent, please contact MBUSA. In any case, this information must not be ignored.

A.7. Production Part Numbers

Although it is a good practice to make use of the part number on the part removed from the vehicle, in some cases the number shown is not a valid spare part number.

The first case is where the part number shown on the part is a "Casting Number", or the general number given to the basic component, not the assembly. One good example is an engine: Cast into the block is what looks like a part number, but instead it is a Casting Number, or the number assigned to the raw block casting. After machining, the same casting can become one of many different part numbers. In other words, the casting number cannot be used to determine the correct part number.

The second case is where the part number shown on the part is a "Production Number", the part number of the item used on the production line. Often, to ease the production process, a part is delivered in a different configuration from that supplied as a spare part.

One example is a lock set: In production, a complete set of locks and keys, with pre-programmed control modules, is delivered to the line. As a spare part, the complete set is not available, but the individual parts are. The control module may be marked with the part number indicating the complete set, rather than the individual component. Another example is a traction control system hydraulic unit: Production receives units pre-filled with fluid, while the spare part version is shipped dry. The production part number is not available as a spare part.

In these cases, attempting to order a casting or production number will be fruitless. What is necessary is that the correct part must be specified using the EPC. In cases where MBUSA must be contacted for assistance, it may be helpful when the casting or production number can be quoted.

However, it should be noted that in some cases, the production part number and the spare part number are the same. In such cases, it becomes trivial to specify the spare part.

A.8. Using Service Tools

The service department also has information tools they use, such as WIS, DAS/SDS, Star TekInfo, Service Bulletins, Wiring Diagrams, etc., and these tools are sometimes useful in finding the correct part. It is time well spent when learning the contents of these tools, and how they are used. Even though you may not desire to repair vehicles, knowing what these tools contain and how they can be used will be valuable.

A.8.1. Workshop Information System (WIS)

WIS is a database of function, diagnosis and repair information for vehicles well back to the 1980s. A technician may come to you with a part number, obtained from WIS. While in some cases, such as special workshop tools, these numbers are valid, in other cases the number provided may not apply to the vehicle in question, or an alternative material can or should be used. In cases where the WIS information conflicts with the EPC, understand that the EPC is generally more up-to-date. Also understand that WIS documents often apply to many country versions, but not necessarily the vehicle in the shop.

A.8.2. Star Diagnosis System (SDS)

The Star Diagnosis System (sometimes referred to as DAS, or Diagnosis Assistance System), is a computer-based system which connects to the vehicle and can provide valuable information from various control modules.

By far the most valuable information obtained from this system is the Control Unit Information, which gives the production part number of the control unit. This is obtained either through a "short test" or from the Information page for the specific system. With the production part number – which is in some cases also the spare part number - you have a better chance of identifying the correct part for the vehicle.

A.8.3. Wiring Diagrams

There are times when the electrical wiring diagrams, used by the service department, can help you find the right part. The first help these diagrams can provide is to help you understand where the item you seek is located in the car, and to what systems it is connected. The second help is the electrical component designation, such as X47/4 for a connector or N3/11 for a control module. In some areas of the EPC, these designations are shown in the part description.

It may be valuable to have the technician assist you in finding the correct wiring diagram, as this is a tool that they use many times each day and their familiarity may save you considerable time.

Once you locate the part you seek on the wiring diagram, take note of what other systems or components are connected to it. This can be particularly helpful when trying to identify a wiring harness.

When specifying a wiring harness repair connector, you need to know the total cross-sectional area (sizes) or the wires to be joined. The size of each wire in the car is found on the wiring diagram. Simply add up the size of each wire that will be joined, and use that value to select the correct wiring repair connector (self-soldering splice). For example, to join three 0.75 mm^2 wires and a 1.5 mm^2 wire, you need a connector capable of handling a total cross section of 3.75 mm^2. Looking in Group 54, we find the correct connector sleeve is a Red one, A002 546 00 41, handling a range from 2.0 to 4.0 mm^2.

A.8.4. Service Bulletins

In the technical bulletins (Dealer Technical Bulletins or DTBs), Recall Campaign bulletins and Service Campaign bulletins published for the service department, specific part numbers and other information may be shown. If the technician is aware there is a bulletin on the vehicle, be sure to look at a copy, you may save considerable time and effort, as well as being sure of selecting the correct part to repair the vehicle.

A regular review of new bulletins can also be helpful in planning for new inventory and being familiar with various work that may occur in the workshop.

A.9 Optical Fiber Cables

Mercedes-Benz uses two optical fiber systems for the transport of audio system information. The older system is known as D2B, and the newer system is known as MOST. Fibers for the different systems appear similar, but are not interchangeable.

MOST fibers are not available individually, instead the optical harness must be ordered.

D2B fibers are available individually in certain lengths. The footnote example below (refer to the image on the next page) will be used to illustrate the process of specifying a replacement D2B optical fiber.

If we look at the first line with a part number, we see the text “A210 440 98 05” in the Wiring Harness column, and the text “!HU - 4895 - CDC - 4895 - HU” in the Interfaces / ES2 column. Similar lines follow beneath this line.

What this means is that the optical cable which is part of wiring harness A210 440 98 05 (which can be specified elsewhere) has fiber optic cables running from the Head Unit (HU, which is the Radio or COMAND unit) to the CD Changer (CDC) and then back to the HU. Between the HU and CDC the length is 4895 mm, and (coincidentally) also 4895 mm from the CDC back to the HU. In this case, you would place an order for part number A000 540 38 08 (as mentioned at the top of the footnote) with ES2 code 4895 - that is, A000 540 38 08 4895.

Note the D2B system always forms a ring, with the optical fiber starting at the HU, going in to the next component, then out and in to the next unit, and so on, going to each component and then back to the HU, forming a ring or loop. The system will not work unless the ring is complete.

To determine which part number to order, first determine between which two components the optical fiber runs, find those two components on the list, then read off the length and use that number as the ES2 code for A000 540 38 08. If necessary, a slightly longer fiber can be used, but never a shorter fiber.

Text footnote | Find footnote

```
035   ===========================================================================================================
035   SPLITTING OF MAIN WIRING HARNESSES, D2B (OPTICAL WAVEGUIDE) AS OF MOD.REF. PACK. 2000 (CODE 800). AS A SP. PT. VAR. BETW. THE
035   FACES THE OPTICAL WAVEGUIDE A 000 540 38 08 HAS TO BE ORDERED  IN ACCORDANCE WITH ITS LENGTH (ES2).
035   ===========================================================================================================
035   ===========================================================================================================
035                                                         ES2 TO A 000 540 38 08
035   ===========================================================================================================
035   WIRING HARNESS      !INTERFACES / ES2
035   -----------------------------------------------------------------------------------------------------------
035   A 210 440 98 05     !HU - 4895 -   CDC   - 4895 -   HU
035   A 210 440 86 05     !HU - 5355 -   CDC   - 5355 -   HU
035   A 210 540 95 33     !HU - 5595 -   CDC   - 6895 -   HU   - 3510 - HU
035   A 210 440 96 08     !HU - 4340 - KOPPLER - 0930 - E-CALL - 5270 - HU
035   A 210 440 97 08     !HU - 5370 - KOPPLER - 0230 - E-CALL - 6925 - HU
035   A 210 440 65 09     !HU - 4340 - KOPPLER - 0445 -   CDC  - 4810 - HU
035   A 210 440 66 09     !HU - 5370 - KOPPLER - 0220 -   CDC  - 5605 - HU
035   A 210 540 91 33     !HU - 4810 -   CDC   - 4810 -   HU
035   A 210 540 93 33     !HU - 4810 -   CDC   - 0930 -   SBS  - 1430 - HU
035   A 210 540 92 33     !HU - 4810 -   CDC   - 1430 -   TEL  - 4810 - HU
035   A 210 440 59 09     !HU - 3630 -   TEL   - 3630 -   HU          - HU
035   A 210 440 60 09     !HU - 5595 -   CDC   - 7000 -   TEL  - 3610 - HU
035   A 210 440 61 09     !HU - 5595 -   CDC   - 0530 -   SBS  - 7000 - TEL    - 3610 - HU
035   A 210 540 97 33     !HU - 5595 -   CDC   - 5595 -   HU
035   A 210 440 62 09     !HU - 5595 -   CDC   - 0530 -   TEL  - 5595 - HU
035   A 210 440 63 09     !HU - 5595 -   CDC   - 0530 -   SBS  - 0530 - TEL    - 5595 - HU
035   A 210 540 99 33     !HU - 5595 -   CDC   - 0365 -   SBS  - 0365 - TEL    - 5595 - HU
035   A 210 540 98 33     !HU - 5595 -   CDC   - 0365 -   TEL  - 5595 - HU
035   A 210 540 96 33     !HU - 5595 -   CDC   - 0365 -   SBS  - 6895 - TEL    - 3510 - HU
035   A 210 540 89 33     !HU - 4810 -   CDC   - 0430 -KOPPLER - 4340 - HU
035   A 210 540 90 33     !HU - 5595 -   CDC   - 0230 -KOPPLER - 5370 - TEL    - 3610 - HU
035   A 210 540 94 33     !HU - 3510 -   TEL   - 3510 -   HU
035   A 210 440 84 09     !HU - 4810 -   CDC   - 4810 -   HU
035   A 210 440 84 09     !HU - 4810 - E-CALL  - 4810 -   HU
035   A 210 440 84 09     !HU - 4810 -  KFZ-I  - 4810 -   HU
035   EXPLANATION: HU = CONTROL PANEL; CDC = CD CHANGER; VCS = VOICE CONTROL SYSTEM CONTROL MODULE; TEL = PHONE, COUPLER = CLUTCH; E
035   -CALL = EMERGENCY CALL SYSTEM; AUT-1 = AUT INTERFACE
```

This section on advanced specifying techniques is a work in progress. Suggestions for topics to be added are welcome. Please contact Dealer Parts Services (for Dealers) or the Customer Assistance Center (for non-Dealers) with your suggestions, or write to:

Mercedes-Benz USA LLC
Parts Technical/EPC
1 Mercedes Drive
Montvale, NJ 07645

Appendix B. Exercises

B.1. Detailed and guided exercises

These exercises assume a basic familiarity with the EPC net application.

B.1.1. Exercise 1: Simple part specification: 560SEL rear brake disk.

1. Enter the VIN WDB 1260391A288401
2. Select Group 42, “Brakes”
3. Select subgroup 045 “Rear Wheel Brake”
4. Click on the Callout number (53) just below the illustration of the rear brake disk.
5. Part Number A 126 423 00 12 should be the only selection on the list. Note that there is a Quantity of 2 used per vehicle.
6. Clear the selected callout using the yellow star icon

B.1.2. Exercise 2: Part Specification with Footnote

1. After completing Exercise 1, click on the callout number (41) for the repair kit for the rear brake caliper. Note the wrench symbol (which denotes a repair kit for the caliper).
2. Two part numbers appear. The second one has a footnote (880). Click on the footnote to open it, if it is not already open.
3. The footnote reads “880 – From Model Year 88”.
4. Your vehicle is a Model year 1987 (you can verify this by finding Code 807 on the data card), so the first part (without a footnote) applies, A001 586 73 43. Note that the vehicle must have a caliper built by Teves (a manufacturer). If not, the vehicle has been modified and the correct caliper should be installed before proceeding.

B.1.3. Exercise 3: Simple specification of an oil filter

1. Erase the previous information by selecting “1. Car”
2. Select Model 129068 (SL 500 USA)
3. Select “Maj. Assy. M – Engine” at the top of the Group listing, since Engine Lubrication is where you find Oil Filters.
4. Note the Engine Baumuster (113.961) is displayed, along with the Engine Groups.
5. Select the Group named “Engine Lubrication”
6. Select the Subgroup named “Oil Filter”
7. Callout 155 is the Oil Filter Kit, and the lines indicate that all of those items are included in the kit. Note that you can also get the oil filter housing, oil cooler, and some hardware and seals as well.
8. After selecting Callout 155, note that the first three numbers are replaced leaving only the fourth as valid: A000 180 26 09

B.1.4. Exercise 4: Part Specification, wiring harness in older vehicle

1. Enter the VIN WDB 1160331A015922
2. Note that no data card is available

3. Select Group 54 “Electrical Equipment and Instruments”. No subgroup menu opens, typical of older vehicles.
4. Using the image number list above the illustration, navigate to Image 4 of 13. (Note only 10 of the 13 images are shown; the blue arrows bring up the others) As you click through the images, note the various items illustrated.
5. In the image, locate and click on callout 147, the Fuel Injection Harness. It is near the front of the engine compartment.
6. The two choices have footnotes. Decide which harness fits this vehicle by referring to the VIN. (A116 540 31 09)

B.1.5. Exercise 5: Part Specification, wiring harness (Advanced)

1. Enter the VIN WDB 2110651A221282
2. Select Group 54 “Electrical Equipment and Instruments”
3. Select Subgroup 545 “Main Cable Harness Frame Floor Unit”. This is the main harness that goes from the front doors to the tail lights.
4. Note how the individual connector pieces are shown, with a connector symbol and a callout. The name associated with the callout will tell you where the connector is used. Also note that the harness itself is drawn much like it is laid out in the vehicle.
5. Select Callout 10, the harness itself. Two parts appear, and from the VINs in the footnotes, both seem to apply to this vehicle. In this case, one harness shows Code 494, and so it becomes the most logical choice. As a demonstration of the powerful filtering being done by the EPCnet using the VIN, let’s try an experiment by doing this the “hard way”: we can learn something, so we’ll use this opportunity to explain advanced navigation.
6. Select the Filter icon at the upper right of the screen (an icon like a blue ‘basket’ with yellow dots), and deselect all the checkboxes (See what the yellow star does!). Click OK.
7. Re-select callout 10. Note that your list contains nearly 50 part numbers. Beneath each part number is the model applicability, in brackets. For example, if [061 065 070] is shown, it indicates that the part applies to Baumusters 211.061, 211.065 and 211.070.
8. Scroll downwards (you may have to also scroll a bit to the right). Note that many parts – half of them, in fact – have an “R” in the “Version” column – this means they are for Right Hand Drive vehicles, and not correct for a USA vehicle. (Note that this does not indicate it is for the Right side of the vehicle!). All those can be eliminated from consideration.
9. Scroll to the top of the list again. About halfway down, note that some of the parts have Code 494, or 489+494, or 494+805, etc.
10. Review the data card by clicking on the Data Card icon at the top. At the lower left, scroll through the Codes. 494 is shown, but 489 is not. Close the data card.
11. Select the part which has all of the following attributes: Has Code 494, but not 489+494, AND applies to ‘065’, AND has a footnote which agrees with the VIN or date.
12. You should have found A211 540 30 09. Of course, we knew that from Step 5 already.

B.1.6. Exercise 6: Subcomponents

1. After completing Exercise 5, deselect all callouts using the yellow star above the illustration. Ensure the Filters are still deactivated, as shown by the red "X" over the Filters icon.
2. Scroll down the parts list and note the listing for callout 30, "Pin Bushing Housing", a plastic electrical connector body. The dot before the description means that "this is a sub-component included with the number listed without a dot above", in this case Callout 10.
3. Note that callout 30 does not apply to only one of the listings for callout 10, but all of them. If there were exceptions, these would be noted with footnotes or codes for Callout 30.
4. Finally, also note that if you order the harness in Callout 10, it will include the connector housing in callout 30 (and many other listed with a dot).

B.1.7. Exercise 7: Part Specification with a color code

1. Enter the VIN 4JG1631571A454312. Note that the VIN prefix 4JG is necessary for the data card to be found, type the VIN with the prefix and it changes automatically.
2. Navigate to Group 68 (Coverings), subgroup 260
3. Look at the parts text listing – none of the callouts in the illustration (for example, 72 in the upper left corner) are on the parts list. In this case, select the next illustration (8 of 9) instead, using the image number list above the illustration, or the right arrow key. Note that the subgroup remains the same – subgroups can have more than one illustration.
4. You're looking for the center console armrest cover – callout 126, top center of the drawing. Select that callout.
5. Note there are three footnotes, 820, 831 and 925 associated with part A163 680 90 39. Click on the blue hyperlink and read them.
6. The first two footnotes (820, 831) relate to a VIN range – our vehicle is within the range (A289560 to A458809), so we're OK. (Note, when the VIN is close to the footnote range, you should verify which version part is correct by inspecting the vehicle – the footnote VIN ranges may not be exact). Our VIN doesn't have an X, so those VIN numbers do not apply.
7. The third footnote, 925 (scroll down to see it) helps us find the ES2 color code for this part.
8. Note that many color footnotes are shown (numbered at the far left), but only one (925, which applies to this part) has hyperlinks for the color codes. (This is not always true, sometimes you need to remember which footnote applies).
9. Above the footnote text area are some boxes (these are also on the data card), at the left is the paint code for this vehicle (040U) and near the right is the interior color code (261A). Going back to the footnote, if we scroll down carefully, following the column with "261" in it, we find that Color Code "9B96" (Anthracite) applies.
10. In this case, the complete part number to order is A163 680 90 39 9B96.

Note that there is more information in the footnotes, and in some cases this additional information may also need to be considered when identifying the correct part.

B.2. *Self Study Exercises:*

The remainder of the exercises are deliberately less detailed, to better help you gauge your own understanding of the specifying process. The answers are given at the end of the exercises.

B.2.1. Exercise 8: Repair kit

1. Enter VIN WDB 2030811F437651
2. Navigate to group 46 and find the subgroup containing the steering rack.
3. Identify the repair kit for the power steering rack valve

B.2.2. Exercise 9: Gasket kit

1. Using the same VIN as Exercise 10, now navigate to the Automatic Transmission aggregate
2. Select the subgroup showing the complete transmission assembly
3. Identify the part number for the transmission gasket kit.

(Hint: Look for the gasket symbol).

B.2.3. Exercise 10: Parts in the gasket Kit

1. Navigate to the <u>sixth</u> image in the Transmission aggregate.
2. Which callouts are also included in the gasket kit?

(Hint: Look for the 'kit of parts' symbol).

B.2.4. Exercise 11: Connectors

For the same VIN as the previous exercises, we need to identify the plastic electrical connector housing for the windshield washer nozzles

1. Navigate to the Electrical Equipment group
2. Locate the subgroup where the heated washer nozzle cable harness is shown
3. Two harnesses are shown, but only one potentially applies. On the harness that does apply, identify the Pin Bushing Housing for the heated Washer Nozzles. Note: There are three nozzles in this model.
4. Also click on another callout associated with the connector – one is a gasket, the other the metal electrical contacts. Note the text "NB" in the Quantity column: This is an abbreviation of the German term for "As Needed". The exact quantity used in the vehicle is not specified.

B.2.5. Exercise 12: Reference to another location

1. Enter VIN WDB 2030401F426255. We will identify the Engine Control Module
2. Navigate to Group 54, subgroup 080. The module shown at the upper right is the Engine control module.
3. Take note of the message on the illustration. Since this vehicle is equipped with Engine M271, it applies to our vehicle.

4. This message is telling us to look in the M271 Engine Aggregate, Group 54, subgroup 400. Navigate there.
5. Select the Engine Control Module.

Hint: Pay attention to the codes.

B.2.6. Exercise 13: Applicability images

1. Using the same VIN as the previous exercise, navigate to Group 54 and select the subgroup for the Alternator.
2. Identify the correct alternator for this vehicle.

B.2.7. Exercise 14: Fuel Filter

· Find the part number for a fuel filter for 124.032-1C-215046.

Hints: Remember to change the VIN prefix to WDB. Look for Fuel System and Fuel Pump.

B.2.8. Exercise 15: Window Spring

· Find the part number for the left front window lifter spring in 124.032-1C-215046

Hint: Front door, window regulator

B.2.9. Exercise 16: Tools

· Find the part number for the screwdriver in the tool kit in 126.039-1A-288398

Hint: Tools

B.2.10. Exercise 17: Tools 2

· Find the part number for the screwdriver in a 190 C (110.001)

No Hints!

B.2.11. Exercise 18: More Gaskets

· Find the part number for the steering box gasket kit for a 116.036 (450 SL 6.9 USA)

Hints: Steering. Green message: Go back to Group and select the "LG Steering" Aggregate, then find the Gasket Kit. Note the symbol used, also note the symbol used for Repair kit (Callout 65)

B.2.12. Exercise 19: Wiper Blade

· Find the part number for the wiper blade (not the rubber insert) for a 1998 E420.

Hint: Electrical System, Pane Wiper.

B.2.13. Exercise 20: Accelerator pedal

· Find the part number for the accelerator pedal for a 1998 E420 (Model 210)

Hint: Control

B.2.14. Exercise 21: Bumper

· Find the part number for a Rear Bumper ("Face Bar" portion) in a 1998 E420 without AMG package, with Brilliant Silver (Code 744) exterior paint.

Hints: Attachment Parts. Read footnotes carefully, and use what is shown.

B.2.15 Exercise 22: Feedback

· Using the Instant feedback button, complete a form reporting an error and print it as a Fax form. Be sure to be viewing the part in question, since the Group, Subgroup, part and VIN information are automatically transferred to the form. Please enter your name, contact phone number, e-mail address and a description of the error you've found. Every report is acknowledged with a phone call or e-mail.

B.3. Answers:

#8: A211 460 08 84

#9: A140 270 65 00

#10: Callouts 14, 17, 25, and 45 (among many) are included in the Gasket kit.

#11: A210 540 20 81 is the connector housing.

#12: A271 153 87 79

#13: A271 154 08 02 or A271 154 09 02 – the lower part of the illustration is valid for Engine M271.

#14: A 002 477 13 01 or A 002 477 17 01

#15 A 000 768 08 34

#16 A 000 581 03 17

#17 A 000 581 03 17, same as the previous exercise!

#18 A 107 460 00 61

#19 A 210 820 00 45. The rubber insert is not available separately.

#20 A 220 300 02 04

#21 A 210 880 19 71 without any ES2 color code. Trick Question: Footnote 420 indicates that it must be painted, and no color code footnote is shown, so only order the base number without color code. If you ordered the part with an ES2 code, it would be rejected.

Appendix C: The Mercedes-Benz Parts Group System

This guide will help explain the Mercedes-Benz spare parts group system. The general contents of each Parts Group are listed. Although it is similar to the group system used for Service, there are some differences.

It is important to note that there are generally three different parts *Catalogs* used for each vehicle's spare parts listing: The Body & Chassis Catalog, the Engine Catalog, and the Transmission Catalog. These additional Catalogs, beyond the Body & Chassis catalog, are known as "Aggregate" Catalogs.

Each Catalog covers certain groups, and some of the Group numbers repeat – for example, Group 27 can be found more than once. The older G Class has separate Chassis and Body Catalogs, as do all commercial vehicles. Some older models have more than two Aggregate Catalogs as well – in all cases, their contents will be clear from the title (such as "Steering, Group LG 46", in Type 123.123, which is a 240D), or in any case can be determined by looking into the Catalog.

C.1 Chassis & Body Groups

Group	Title	Contents
21	Major Assembly Detachable Body Components	Fasteners, coverings, brackets, and V-belts for Aggregates
24	Engine Suspension	Engine mounts, brackets and hardware
25	Clutch	Clutch plate, lining, hardware
26	Gearshift Mechanism	Manual gearshift mechanism, shifter handle, shifter boot
27	Automatic Transmission	Floor shifter mechanism, shifter handle, oil filler pipe
28	Transfer Case	Assembly, and all components for assembly
29	Pedal Assembly	Brake and clutch pedals, brackets and hardware
30	Control	Accelerator pedal assembly
31	Frame, Trailer Coupling	Chassis Frame, brackets & mountings, Trailer Hitch & Harness
32	Springs, Suspension & Hydraulic System	Shocks, springs, anti-roll bars, oil lines, brackets and mountings
33	Front Axle	Axle, axle shafts, steering knuckles, hubs, bearings, control arms, headlamp adjusters
35	Rear Axle	Axle, axle shafts, subframe, bearings, hubs, control arms, headlamp adjusters
40	Wheels	Wheels, lug bolts, trim, balance weights.
41	Propeller Shaft	Driveshaft, flexible joint
42	Brakes	Pads, discs, calipers, booster, master cylinder, ABS control unit, lines, parking brake pedal and cables
46	Steering	Rack, tie rods, lines, power steering pump, columns, wheel, airbag, angle sensor
47	Fuel System	Tank, pumps, sending units, filter, charcoal filter, and shut off valve
49	Exhaust System	Exhaust pipes, manifolds, catalytic converters, mufflers, heat shields and mounting hardware

50	Radiator	Radiator, fan, reservoir tank, hoses, transmission cooler lines, intercooler
52	Chassis Sheet Metal/Air Intake	Engine encapsulation shield, air intake hoses and engine covers, Engine air filter in some cases
54	Electrical Equipment & Accessories, Chassis	Wiring harnesses, control units, battery, alternator, instrument cluster, steering column switches (combi-switch, cruise control, ignition switch), light switch, sensors, loom ties and cable mounting parts
58	Tools & Accessories	Tools, Tool accessories, and labels
60	Shell (Body)	Complete body shell, expansion plugs and grommets
61	Substructure	Floor sheet metal, wheelhouse, spare wheel well
62	Front End, Front Wall	Frame, radiator support, firewall, support under instrument cluster
63	Side Panels	Sheet metal and insulation for A, B, and C pillars, rear panels and gas flap
64	Rear End	Sheet metal for hatshelf and rear inside trunk
65	Roof	Roof sheet metal
67	Windows	Front and rear windshields, seals, installation kits
68	Panelling	Dashboard, center console, floor lining, door sill moldings, hatshelf, insulation, roller blind, airbag
69	Panelling & Lining	Sun visors, rear view mirror, head curtain airbags, interior trim, headliner, interior trunk trim, outside door and rocker moldings
72	Front Doors	Door, paneling and trim, glass, side mirrors, lock and keys, window regulator, handle and latch
73	Rear Doors	Door, paneling and trim, glass, window regulator, handle and latch
74	Rear Panel Door	Door, paneling and trim, glass, emblems, handle and latch
75	Rear Lid	Trunk lid, trim and latches
77	Soft Top	Folding top assy, cover, frame, brackets and mounting
78	Sliding Roof	Sliding roof assembly, frame, glass, motor
79	Hard top	Hard top assembly, glass, seals, trim
80	Vacuum System	Pumps, lines, connectors, elements, folding top hydraulics and lines, brackets and mounting parts
81	Appointments	Sun visors, rearview mirrors, ashtrays, grip rails
82	Electrical System, Body	Switches, windshield wipers, exterior body lights, SRS, audio, tele-aid, antennas
83	Heating and ventilation	Heater case, blower, evaporator, A/C control unit, ducts, compressor, condenser, hoses
84	Trunks and Cases	Roof rack, trunk strap, battery in luggage compartment
86	Washer System	Reservoir, sensor, lines, nozzles
88	Attachment Parts	Bumpers, fenders, hood, grille, sensors
89	Accessories	Jack, wheel chock, lockset, burglary alarm
91	Front Seats	Covers, cushions, headrest, motors, seat adjusters, heaters, seat belts

92	Rear Seats	Covers, cushions, headrest, motors, seat adjusters, heaters, seat belts,
93	Center Seat / Bench	Covers, cushions, seat belts, trim, brackets and mounting parts
97	Seat Accessories	Seat belts, front seat folding armrest, seat heaters
98	General Parts	Material
99	Special Internal Fittings	Orthopedic backrest, seat heating, miscellaneous,

C.2 Engine Groups

Group	Title	Contents
M 01	Engine Housing	Block, cylinders, timing case, oil pan, valve covers
M 03	Moving Parts	Crankshaft, flywheel, pistons
M 05	Timing	Camshaft, timing chain, valves
M 07	Injection	Fuel injectors, distributors, lines, linkage
M 09	Air Cleaner	Air filters, covers, compressors, hoses
M 13	Power Steering Pump, Refrigerant Compressor	Power steering pump, compressor, hoses, belts
M 14	Intake & Exhaust Manifolds	Intake & Exhaust manifolds, EGR valve, air pumps, vacuum lines
M 15	Electrical Equipment, Engine	Starter motors, alternators, ignition system, sensors
M 18	Engine Lubrication	Oil pumps, oil filters
M 20	Engine Cooling System	Water pumps, fans, pulleys, belt tightener
M 22	Engine Suspension	Engine brackets and mounting parts, see also Group 24
M 23	Hydraulic System	Generally empty, see Group M13
M 54	Engine Cable Harnesses	Engine harness and connectors, battery cable

C.3 Other Groups

Group	Title	Contents
GM26	Manual Transmission	Complete assembly and parts
GA27	Automatic Transmission	Complete assembly, torque converter, and parts
LG46	Steering Box	Complete assembly and parts

Parts Specification Guide
Version 7 May 2007
EPCnet 1.17

Mercedes-Benz USA, LLC
Parts Assistance Center, Parts Technical
1 Mercedes Drive, Montvale, NJ 07645
(877) PARTSMB